

Airline scheduling optimization: literature review and a discussion of modelling methodologies[†]

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The rapid development of civil aviation in recent decades has not only led to increasing competition among airlines, but also to the rise of irregularities, with challenges concerning the improvement of regulations and schedules under the maximization of profitability. Consequently, the airline scheduling optimization problem has received significant research interest as the foundation for an efficient deployment of airline resources and meeting market demand under complex operational requirements. In this paper, we dissect fundamental airline scheduling problems by reviewing thirteen representative mathematical models for schedule design, fleet assignment, aircraft routing, crew scheduling subproblems and their potential for integration. In contrast to existing review studies on airline scheduling problems, our main contribution lies in the introduction of state-of-the-art mathematical models with a specific focus on integration and robustness. In addition, we highlight a set of promising, yet challenging, directions for future research in this domain.

Keywords: airline scheduling; fleet assignment; aircraft routing; crew scheduling; robustness

1. Introduction

Escalating demand for mobility and social connections has led to the continuous development of the aviation industry over recent decades. In 2019, about 4.5 billion passengers were carried by means of scheduled air transportation, marking a notable 3.6 per cent increase compared to the previous year (ICAO 2020). Although the COVID-19 pandemic confronted the aviation industry with unprecedented challenges (Sun et al. 2021, 2022; Standfuss et al. 2023; Xu et al. 2023), it was predicted that global passenger traffic would rapidly rebound to pre-pandemic levels by the end of 2023 (IATA 2023a; Sun et al. 2023a, b). Increased air traffic flow in the future will necessarily lead to tighter requirements for the safe and efficient operation of aircraft, crew, and passenger services. On the other hand, airlines are typically confronted with substantial expenses associated with fuel, labour, maintenance, and so forth. Further incorporating stricter regulations and facing uncertainties in demand/operational disruptions (e.g. flight delays and airport closure) leads to a rather low net profit margin (Ding et al. 2023; IATA 2023b). Therefore, it is of great importance to improve the profitability of schedules and reduce the related operational cost by using effective and efficient airline scheduling optimization techniques.

From the perspective of airlines, profitable management of their two crucial assets, i.e. aircraft and cabin crew, refers to schedule planning and is usually decomposed into four sequential subproblems, namely the schedule design problem (SDP), the fleet assignment problem (FAP), the aircraft routing problem (ARP), and the crew scheduling problem (CSP) because of different

planning horizons and great complexity (all subproblems are NP-hard). Since schedule design is more flexible given airlines' individual preferences on marketing and empirical strategies, it is usually combined with FAP and ARP in publications and no well-acknowledged model particularly for SDP is available (Zhou et al. 2020). For an overview of the literature on the airline scheduling problem and the specific subproblems, the reader is referred to several comprehensive reviews in this area (Sherali et al. 2006; Eltoukhy et al. 2017; Zhou et al. 2020; Wen et al. 2021). These surveys provide a thorough summary of the proposed models and solution techniques in the literature, while often neglecting the specific modelling architectures and challenges inherent to integrated and robust scheduling problems.

Specifically, from an optimality viewpoint, sequentially solving scheduling subproblems ignores the dependence between different operating resources to some extent and leads to suboptimal or even infeasible solutions with degraded performance (Xu et al. 2024). For instance, the solution of the fleet assignment subproblem may lead to inappropriate allocation of maintenance resources and is likely to incur routing solutions with high maintenance costs. To address these limitations, integrated optimization has been the subject of increasing research interest recently and a cost saving of around 2 per cent has been reported in the literature (Papadakos 2009). From a practical standpoint, flight schedules frequently suffer from operational irregularities and—at design time—uncertain inputs, including flight delays, convective weather, and fluctuating passenger demand. Consequently, the highly optimized schedules of aircraft, crews, and

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Table 1. Characteristics of the selected models from M1 to M13.

Category	Problem	Literature	Model	Network	Horizon	Characteristic
Fleet assignment	FAP	Barnhart et al. (2002)	M1	TSN	Daily	Reduced model size, itinerary-based demand
		Abara (1989)	M2	CN	Daily	Leg-based demand, applicable to through flight
Aircraft routing	ARP	Liang and Chaovalitwongse (2013)	M3	TSN	Weekly	Max elapsed days before maintenance, aggregated overnight/maintenance arcs
		Başdere and Bilge (2014)	M4	CN	Operational	Remaining time before maintenance
		Barnhart et al. (1998)	M5	RN	Daily	Max elapsed days before maintenance, augmented string
Crew scheduling	CPP	Saddoune et al. (2013)	M6	RN	Dated	Connection network-based pairing generation
	CPP+CRP	Zeighami et al. (2020)	M7	RN	Monthly	Pairing and rostering generation with consistency
Integrated scheduling	SDP+FAP	Lohatepanont and Barnhart (2004)	M8	TSN	Daily	Optional flight selection
	SDP+FAP+ARP	Sherali et al. (2013)	M9	CN	Daily	Optional flight selection and basic routing without maintenance
	ARP+CPP	Ruther et al. (2016)	M10	RN	Operational	Aircraft route and crew pairing inconsistency reduction
Robust scheduling	FAP+CPP	Sandhu and Klabjan (2007)	M11	TSN+RN	Weekly	Plane-count constraint based on essential ground arcs
	SDP+FAP	Cadarso and Celis (2017)	M12	TSN	Daily	Demand fluctuation via stochastic programming
	CPP	Antunes et al. (2019)	M13	RN	Daily	Budge uncertainty set for flight root delays

FAP: fleet assignment problem; ARP: aircraft routing problem; CPP: crew pairing problem; CRP: crew rostering problem; SDP: schedule design problem; RN: route network; CN: connection network; TSN: time-space network.

passengers can cause delay propagation, flight cancellation, additional crew deadhead, and passenger overbooking effects, leading to low operation efficiency. To mitigate such risks, a proactive approach has been developed to develop robust schedules and reduce the overall probability of disruptions (Khaled et al., 2018b).

The remainder of this paper is organized as follows. Section 2 provides the essential notations, examples, and commonly used network structures shared across standard models. Section 3 illustrates the fleet assignment problem concerning its representative model formulations. Following that, the research progress into aircraft routing and crew scheduling (including pairing and rostering) is presented in Sections 4 and 5, respectively. Section 6 surveys the integrated problems that jointly solve two or more subproblems. In Section 7, robust models for stochastic demand and delay propagations are reviewed. Finally, Section 8 concludes our study and provides a set of recommendations for future work. In summary, while Section 2 and Section 3 build the foundation of this review, the major novel contribution can be seen in the parts covered by Section 4–Section 8.

2. Preliminaries

2.1 Overview and general nomenclature

Airline scheduling is a challenging problem involving decisions on the allocation of aircraft, crew, and flight resources to meet passenger demand and accommodate complex operating restrictions, such as aircraft maintenance. Traditionally, the following subproblems, SDP, FAP, ARP, and CSP, are sequentially addressed to not only reduce the computational burden but align with different planning horizons (referring to strategic or operational planning) as well. Alternatively, there has been a growing popularity in recent decades in integrating the decisions across two or more subproblems, aiming to reduce the inconsistency and enhance

Table 2. Common sets and parameters for mathematical models.

Sets	
F	Set of flights, $f \in F$.
K	Set of fleet types, $k \in K$.
A	Set of aircraft, $a \in A$.
R	Set of aircraft/fleet routes, $r \in R$.
I	Set of passenger itineraries, $i \in I$.
P	Set of crew pairings, $p \in P$.
AP	Set of airports, $ap \in AP$.
Parameters	
D_i	Passenger demand for itinerary i .
S_k/S_a	Capacity of fleet k /aircraft a .
N_k	Number of available aircraft of fleet k .
c_{fk}/c_{fa}	Operation cost of assigning flight f to fleet k /aircraft a .
fare_i	Airfare of passenger itinerary i .

the airline profits, albeit with an expense of significantly higher computational complexity.

Based on the characteristics of the tackled problems, model formulation, and underlying network structures, thirteen representative optimization models (see Table 1) are selected in this paper to better underscore the differences and similarities of the airline scheduling variants. Specifically, the Problem and Network columns in the table represent the covered subproblems and underlying networks of the selected models, respectively. The Horizon column indicates the planning horizon of the generated schedules (i.e. daily/weekly repeated or operational).

Before formally stating the specific mathematical formulation, the nomenclature pertaining to commonly used notation is defined in Table 2 and will be employed in subsequent sections.

Table 3. Flight timetable of the illustrative example.

Flight	Depap	Arrap	Deptime	Arrtime
F1	Airport A	Airport B	08:00	09:30
F2	Airport B	Airport C	10:10	11:40
F2	Airport C	Airport A	12:20	15:20
F4	Airport A	Airport B	07:30	09:00
F5	Airport B	Airport A	10:15	11:45
F6	Airport A	Airport C	13:00	16:00
F7	Airport C	Airport A	16:30	19:30

Furthermore, some critical definitions are given as follows:

- *Flight*: a non-stop trip executed by one aircraft from one airport to another at a specific departure and arrival time.
- *Itinerary*: a sequence of flights with the first flight departing from the origin airport and the last flight arriving at the destination airport. Different departing times and passenger classes can also be included. Both non-stop and one-stop itineraries are considered in this context.
- *Maintenance station*: an airport that is capable of performing maintenance operations for aircraft after flying a certain amount of time.
- *Spill and recapture*: spill occurs when the assigned aircraft for one flight cannot accommodate all passenger demand. Spilled passengers may be recaptured by booking alternative flights operated by the same airline.

2.2 Illustrative example

A simple flight schedule consisting of six flights daily is created and shown in Table 3 with information on flight number, departure/arrival airport (Depap/Arrap), and departure/arrival time (Deptime/Arrtime). The flight block time for airport pairs (A, B) and (B, C) is assumed to be 90 min and that for pair (A, C) is set to 180 min; the minimum turnaround time between two consecutively executed flights is 30 min by default.

In this study, we present an airport network (see Fig. 1a) based on the flight schedule and illustrate several commonly used network structures in the airline scheduling optimization field, namely the time-space network and connection network. Additionally, we also describe the route network to match with publications using set partition formulation in their mathematical models.

2.2.1 Time-space network

In the time-space network, each airport is associated with a timeline featuring a sequence of departure and arrival events in increasing order of event time. Accordingly, each vertex in the network corresponds to a departure or arrival event in the flight schedule. Three types of edges, *flight edges*, *ground edges*, and *overnight edges*, are introduced to represent the aircraft movement. Flight edges pertain to flight segments, while ground edges represent the number of aircraft on the ground at the same airport for a specific time period; the time for arrival events equals the arrival time plus the minimum turnaround time for building flight connection opportunities. In order to build a repeated schedule, overnight edges are introduced, which connect the last event with the first event of every airport to build a cyclic network. In Fig. 1b, a time-space network is built based on the flight schedule in Table 3. The spatial and temporal information

is displayed vertically and horizontally. The flight edges and overnight edges are indicated by solid arrows and dashed arrows, respectively. Based on this network, flight connections can be distinguished in the same timeline, such as (F1, F2), (F4, F5).

2.2.2 Connection network

The connection network is built on flight connections where each vertex represents a flight (or flight departure/arrival event) and each edge denotes a possible connection between two flights i , j given that the arrival airport of i is the same as the departure airport of j and the connection time is larger than the minimum turnaround time. To build repeated schedules, originating and terminating edges are created between flight vertices and airport source/sink vertices for cases when aircraft start and end at different airports with overnight edges to build a cyclic graph. In Fig. 1c, solid arrows are edges between flights and edges between airport vertices are implicitly indicated with grey timelines. The first two figures provide clear demonstrations of their distinctions. While a time-space network features a smaller size and lower complexity than the equivalent connection network because flight connections are aggregated through ground edges, the absence of connection information limits its use concerning through flights and delay propagation. Therefore, both networks are widely used for different purposes.

2.2.3 Route network

In this part, we introduce a route network derived from either the time-space network or the connection network. Set partition formulation is a popular modelling concept for ARP and CSP where the decision variables are modelled based on sequences of flights rather than individual flights or flight connections. In this way, column-generation-based solution methodologies can be applied for both better scalability and linear relaxation. In Fig. 1d, we provide one example of this network. Each edge of this network represents an aircraft route or a crew route from the source airport to the sink airport. In route R3, flights F4, F5, and F6 are included with airports A, B, A, and C being visited sequentially. Therefore, many operational constraints can be embedded in the construction of route variables to simplify the model formulation.

3. Fleet assignment problem

As a critical determinant of airline revenue, FAP involves assigning different aircraft types to the scheduled flights given different capacities, operation costs, and other factors for profit maximization purposes. From the perspective of the underlying network, the mathematical formulation can be categorized into two types: *time-space network-based models* and *connection network-based models*.

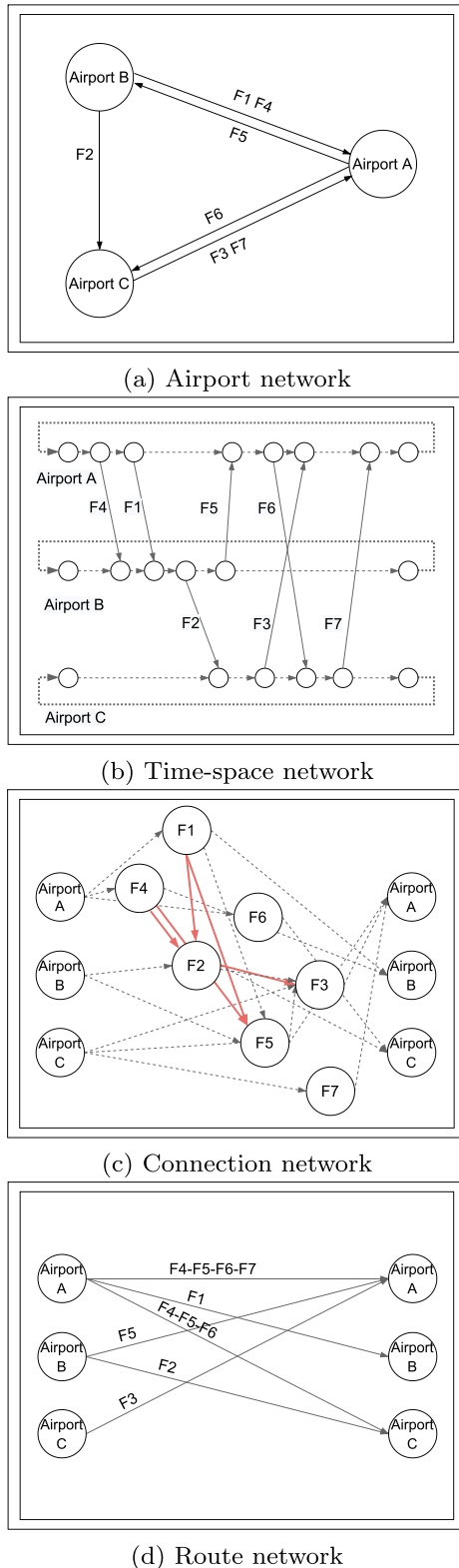


Figure 1 Network representation of the example flight schedule.

In this section, these two basic models and extension are discussed and summarized in Table 4.

3.1 Time-space network-based model

Compared to the connection network, the time-space network was proposed and utilized in Berge and Hopperstad (1993);

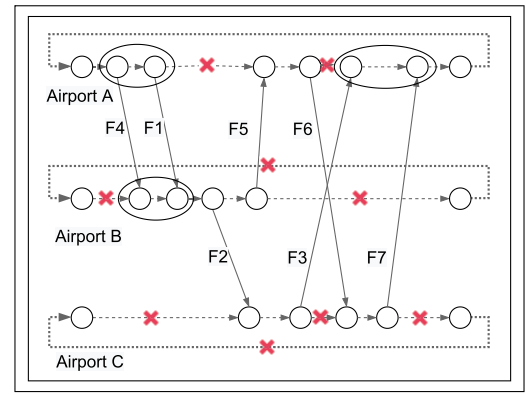


Figure 2 Preprocessing techniques for the time-space network.

Hane et al. (1995) with fewer arcs, albeit at a loss of flight connection information, which limits its application to tail-specific routing. Three types of edges are constructed, namely flight edges, ground edges, and wrap-around edges. To reduce computational difficulties, Hane et al. (1995) further introduced three preprocessing techniques, *node aggregation*, *isolation*, and *eliminate missed connections*, aimed at reducing the number of vertices and edges in the time-space network. The node aggregation and isolation techniques for the example time-space network are shown in Fig. 2 as ellipses and red crosses, respectively.

Based on this basic model, several extensions have been proposed to tackle the flexible departure times and network effects. To accommodate variable departure times, Rexing et al. (2000) presented a mathematical model where the departure time of each flight is discretized within time windows as flight copies, thereby increasing the flight connection opportunities. Because the inclusion of several copies increases the problem size, an iterative solution technique was developed consisting of a master problem and a subproblem for superoptimality and feasibility, respectively. Another extension of the basic FAP was the combination of network effects where passenger demand is dependent on the available seats and not identical concerning resultant revenue/resource consumption. In Barnhart et al. (2002), a passenger mix model was proposed to minimize the loss of revenue due to spills. Passenger recapture rates were evaluated using the quality of service index. Furthermore, the network effects and the stochastic nature of demand were addressed by Jacobs et al. (2008) through an integrated fleet assignment and revenue management model, employing linear approximations to the total revenue function. Subsequently, Pilla et al. (2012) used multivariate adaptive regression splines cutting plans to solve the two-stage stochastic programming model with respect to demand uncertainty. Compared to the traditionally adopted L-shaped method, a significant amount of computation time saving was achieved. To better integrate revenue management decisions in FAP, Barnhart et al. (2009) proposed a subnetwork fleet assignment model that provides an accurate approximation to the itinerary-based fleet assignment model with good computation efficiency. A solution approach was developed for a reformulated model, partitioning fare products and enumerating decision variables to save computation time. Focusing on the high uncertainty of demand, Liu et al. (2018) presented a risk-averse two-stage stochastic programming model for FAP based on the time-space network to maximize total profit under multiple demand realization scenarios. The corresponding solution technique relies on the sample average

Table 4. Summary of the literature on the fleet assignment problem.

Literature	Network	Time horizon	Recapture	Method	Fleet	Flight	Time
Abara (1989)	CN	Daily		B&B	4	400	3,600 s
Berge and Hopperstad (1993)	TSN	Weekly		Sequential minimum cost flow	3	8,310	221.3 s
Subramanian et al. (1994)	TSN	Daily	✓	Cutting-edge	10	2,500	10,800 s
Hane et al. (1995)	TSN	Daily		B&B	11	2,500	883.2 s
Rushmeier and Kontogiorgis (1997)	CN	Daily		B&B	8	1,610	7,200 s
Ioachim et al. (1999)	CN	Weekly		DW, CG	–	106	4,186.4 s
Rexing et al. (2000)	TSN	Daily		Iterative heuristic	11	2,037	402.6 s
Barnhart et al. (2002)	TSN	Daily	✓	CG, RG	9	1,888	3,076.22 s
Rosenberger et al. (2004)	CN	Daily		First-out policy, B&B	9	–	–
Bélanger et al. (2006a)	TSN	Daily		DW, CG	7	412	22,080 s
Bélanger et al. (2006b)	TSN	Weekly		Two-phase heuristic	10	782	23,229 s
Smith and Johnson (2006)	TSN	Daily/weekly		CG	19	4,182	1,255.80 s
Ahuja et al. (2007)	CN	Daily		VLNS	13	1,609	24 s
Jacobs et al. (2008)	TSN	Daily	✓	Iterative algorithm	–	2,200	–
Pilla et al. (2008)	TSN	Daily		Statistical experiments	7	2,358	2.9 d
Barnhart et al. (2009)	TSN	Daily		Partition heuristic	9	2,044	3,838 s
Pilla et al. (2012)	TSN	Weekly		BD	7	2,358	5.8 d
Liu et al. (2018)	TSN	Daily		SAA	6	72	57,120 s

CN: connection network; TSN: time-space network; B&B: branch and bound; BD: Benders decomposition; CG: column generation; DW: Dantzig-Wolfe decomposition; RG: row generation; VLNS: very large neighbourhood search; SAA: sample average approximation; –: not reported.

algorithm and is validated through small real-world instances of a Chinese legacy carrier.

In Barnhart et al. (2002), a passenger mix model was proposed to minimize the loss of revenue due to spills. Passenger recapture rates were evaluated using the quality of service index. The corresponding mathematical model is presented in Constraints (1)–(6) with the notation provided in Table 5:

$$(M1) \quad \min -R + \sum_{f \in F} \sum_{k \in K} c_{k,f} x_{k,f} + \sum_{i_1 \in I} \sum_{i_2 \in I} (\text{fare}_{i_1} - b_{i_1}^{i_2} \text{fare}_{i_2}) t_{i_1}^{i_2} \quad (1)$$

$$\text{s.t.} \quad \sum_{k \in K} x_{k,f} = 1 \quad \forall f \in F \quad (2)$$

$$y_{k,ap,t^-} + \sum_{f \in FI(k,ap,t)} x_{k,f} - y_{k,ap,t^+} - \sum_{f \in FO(k,ap,t)} x_{k,t} = 0 \quad \forall \{k, ap, t\} \in V \quad (3)$$

$$\sum_{ap \in AP} y_{k,ap,t_j^+} + \sum_{f \in CF(k)} x_{k,f} \leq N_k \quad \forall k \in K \quad (4)$$

$$\sum_{k \in K} S_k x_{k,f} + \sum_{i_1 \in I} \sum_{i_2 \in I} \delta_f^{i_1} t_{i_1}^{i_2} - \sum_{i_1 \in I} \sum_{i_2 \in I} \delta_f^{i_2} b_{i_1}^{i_2} t_{i_1}^{i_2} \geq \sum_{i \in I} \delta_f^i D_i \quad f \in F \quad (5)$$

$$\sum_{i_2 \in I_{i_1}} t_{i_1}^{i_2} \leq D_{i_1} \quad \forall i_1 \in I. \quad (6)$$

In this model, the objective function (1) consists of the operational cost and the loss of profit where the initial unconstrained revenue is computed as $R = \sum_{i \in I} \text{fare}_i D_i$. Constraints (2) are flight cover constraints while flow conservation constraints (3) ensure that aircraft going into an airport at a particular time must leave that airport at some later time. Constraints (4) are count constraints that ensure that the available number of aircraft of each type is utilized. This is achieved by selecting an arbitrary count time and enumerating all edges (flight, ground, and

Table 5. Additional notation for the time-space network-based fleet assignment model M1.

Sets	
I_i	The set of candidate redirected itineraries of itinerary i .
T	The sorted set of all event times in increasing order, $t_j \in T$.
V	The set of vertices in the time-space network, $\{k, ap, t\} \in V$.
$FI(k, ap, t)$	The set of inbound flights to vertex $\{k, ap, t\}$.
$FO(k, ap, t)$	The set of outbound flights to vertex $\{k, ap, t\}$.
$CF(k)$	The set of flights that pass the count time when flown by fleet k .
Parameters	
R	Initial unconstrained revenue.
$b_{i_1}^{i_2}$	Recapture rate from itinerary i_1 to i_2 .
δ_f^i	1 if itinerary i includes flight f , 0 otherwise.
$c_{k,f}$	Cost of operating flight f with fleet k .
Decision variables	
$x_{k,f}$	1 if flight f is assigned to fleet k , 0 otherwise.
$t_{i_1}^{i_2}$	Number of passengers requesting itinerary i_1 but who are redirected to itinerary i_2 .
y_{k,o,t_j^+}	The number of aircraft in fleet k on the ground at airport immediately after t_j .
y_{k,o,t_j^-}	The number of aircraft in fleet k on the ground at airport immediately before t_j .

wrap-around edges) passing through it. The fleet capacity is modelled with Constraints (5), where the second and third terms can be interpreted as the number of passengers spilling from itinerary i_1 and recaptured by itinerary i_2 . Constraints (6) are the demand constraints that restrict the total number of passengers spilling from itinerary i_1 to the unconstrained demand for itinerary i_1 .

3.2 Connection network-based models

The connection network is widely used to depict the spatial and temporal relationships within airline operations. As the pioneering work that builds FAP on the connection network, Abara (1989) introduced a basic model formulation composed of three index binary variables x_{ijk} to determine if connection (i, j) can be carried

out consecutively by fleet k . Using a similar model formulation, Rushmeier and Kontogiorgis (1997) improved computational performance by introducing preprocessing techniques where flight operation connections at airports are partitioned into different connecting complexes to avoid infeasible connections outside one complex. While the daily schedule is a widely adopted assumption in the literature, FAP for the weekly schedule is studied in Ioachim et al. (1999) to address the problem that the schedule may vary over the days. Consequently, additional decision variables for departure time are introduced in their model and schedule synchronization constraints are also included to ensure that the departure time of a flight number remains the same during the planning horizon. Regarding the solution algorithm, they adopt the Dantzig–Wolfe decomposition to the multi-commodity flow model with specific branching strategies for optimal solutions.

To alleviate the cascading effects of cancellation and delays, Rosenberger et al. (2004) presented a model for robust FAP where rotation cycles are created with few hub airports being visited. To be specific, the aircraft route *cancellation cycle* is defined as a string starting and ending at the same airport. Otherwise, the route is defined as an *acyclic string*. Hence, a flight schedule with more cancellation cycles and fewer acyclic strings is proven to be robust against unexpected disruptions and alternative models are formulated to constrain the number of acyclic strings. Furthermore, the basic fleet assignment model was extended in the work of Ahuja et al. (2007) to optimize through revenue as well. Due to its high complexity, direct use of a commercial solver is hindered. Accordingly, they proposed a very large neighbourhood search algorithm based on the connection network. In particular, six types of arcs are defined in an improvement graph to express local search operators; e.g. fleet type swap and negative cost valid cycles are found from the improvement graph iteratively to reduce the total operational cost.

With the additional notation provided in Table 5, the mathematical formulation of the fleet assignment model from Abara (1989) is presented in Constraints (7)–(11):

$$(M2) \max \sum_{f_1 \in F} \sum_{f_2 \in F} \sum_{k \in K} p_{f_2 k} x_{f_1 f_2 k} - \sum_{f \in F} \sum_{k \in K} c_{fk} x_{0fk} \quad (7)$$

$$\sum_{f_1 \in F \cup \{0\}} \sum_{k \in K} x_{f_1 f_2 k} = 1 \quad \forall f_2 \in F \quad (8)$$

$$\sum_{f_1 \in F \cup \{0\}} x_{f_1 f_2 k} - \sum_{f_2 \in F \cup \{0\}} x_{f_2 f_1 k} = 0 \quad \forall f_2 \in F, k \in K \quad (9)$$

$$\sum_{f \in F_{ap}^D} x_{0fk} - \sum_{f \in F_{ap}^A} x_{f0k} = 0 \quad \forall ap \in AP, k \in K \quad (10)$$

$$\sum_{f \in F} x_{0fk} \leq N_k \quad \forall k \in K \quad (11)$$

The objective function (7) aims to maximize profits while minimizing the operational costs. To represent the initial flights executed by different aircraft, we introduce 0 as the artificial source vertex, where $x_{0fk} = 1$ indicates the assignment of fleet k to flight f . Constraints (8) ensure that every flight is covered once. Constraints (9) maintain flow balance on every flight. Constraints (10) help to build a repeatable schedule by restricting the number of aircraft at the beginning of the day to that at the end of the day for each airport. The aircraft count constraints are given in Constraints (11). Due to the large cardinality of decision variables $x_{f_1 f_2 k}$ associated with the size of the connection network, this model is more challenging to solve in contrast to an equivalent time-space network-based model.

Table 6. Additional notation for the connection network-based fleet assignment model M2.

Sets	
F_{ap}^D	Set of flights departing from airport ap .
F_{ap}^A	Set of flights arriving at airport ap .
Parameters	
p_{fk}	Profit of executing leg f using fleet k .
c_{fk}	Operating cost of flight f with fleet k .
Decision variables	
$x_{f_1 f_2 k}$	1 if connection between f_1 and f_2 is executed by fleet k , 0 otherwise.
x_{0fk}	1 if originating flight f is executed by fleet k .
x_{f0k}	1 if terminating flight f is executed by fleet k .

In summary, over recent decades, there have been significant efforts devoted to FAP in the literature and recent research has shown great interest in tackling more realistic issues considering weekly schedules, itinerary-based demand, passenger dynamics, and uncertainty regarding demand and operation perturbations. These new features result in an inevitable increase of computational complexity, thereby posing challenges to its applicability in the airline industry. Therefore, developing scalable and accurate solution techniques remains a promising research direction for more complicated problems. Besides, it is also crucial to strike an appropriate trade-off between computation efficiency and solution robustness in order to avoid potential demand overestimation or underestimation.

4. Aircraft routing problem

Given the assignment of fleets from FAP, ARP aims to build individual routes flown by each aircraft. A feasible aircraft route must satisfy aircraft maintenance requirements mandated by aviation authorities and airlines. Common scheduled maintenance tasks include a Type A check (every 400–600 flight hours), Type B check (every six to eight months), Type C check (every two years), and Type D check (every four years). In practice, the exact required intervals for different tasks vary according to fleets, aircraft age, etc., and a typical Type A check is often split into several small packages called a phased-A check to meet up with tight flight schedules. Type B, Type C, and Type D tasks are less frequently incorporated in the literature due to their long intervals between adjacent maintenance instances. On the basis of the modelling choices, ARP is categorized into three classes: *time-space network*, *connection network*, and *route network*. The summarized publications are shown in Table 7. Because aircraft operation costs are considered in the fleet assignment problem, maintenance costs are required for the models presented in this section.

4.1 Time-space network-based models

Since airlines usually operate maintenance checks during the night to maximize the aircraft utilization rate, especially for domestic operations, compact time-space network-based models are proposed based on a simplification of maintenance requirements. Only maximum accumulated calendar days are considered and flight connection information is discarded due to the aggregation of ground arcs as presented in Liang et al. (2011); Liang and Chaovaitwongse (2013). The work by Khaled et al. (2018a) develops a novel model with higher complexity. In order to consider cumulative maintenance requirements, big-M constraints and additional variables are introduced. Aircraft initial positions are also taken into account.

Table 7. Summary of the literature on the aircraft routing problem.

Literature	RN	CN	TSN	Objective	Method	A/C	Flight	Time
Clarke et al. (1997)		✓		Max through value	LR	–	–	39.3 s
Barnhart et al. (1998)	✓			Min operation cost	B&P	89	1,124	16,200 s
Gopalan and Talluri (1998)	✓			Feasibility	Polynomial-time heuristic	12	33	–
Talluri (1998)	✓			Feasibility	Eulerian routing	–	–	–
Sriram and Haghani (2003)		✓		Min maintenance and reassignment cost	Depth-first search, random search	8	58	300 s
Sarac et al. (2006)	✓			Min unused flying hours	B&P	32	175	1,940 s
Lan et al. (2006)	✓			Min propagated delays	B&P	–	278	13 s
Burke et al. (2010)			✓	Max reliability and flexibility	GA	–	504	98 s
Liang et al. (2011)			✓	Min undesirable connections	B&B	70	352	15 s
Liang and Chaovalitwongse (2013)			✓	Max through value	Diving heuristic	333	2,086	881 s
Haouari et al. (2013)		✓		Feasibility	B&B, reformulation	138	344	10.33 s
Başdere and Bilge (2014)		✓		Min unused flying hours	Compressed annealing	8	354	8,513.57 s
Froyland et al. (2014)	✓			Min recovery and deviation cost	BD, CG	10	53	4,000 s
Liang et al. (2015)	✓			Minimize expected propagated delays	CG	333	6,072	10,059 s
Khaled et al. (2018a)			✓	Min operation cost	B&B	40	1,494	10,800 s
Safaei and Jardine (2018)		✓		Min operation cost	Iterative solution algorithm	18	772	–
Yan and Kung (2018)	✓			Min-max propagated delays	CG, RG	23	117	446.10 s
Maher et al. (2018)	✓			Min maintenance violations	B&P	526	3,370	76.8 s
Cui et al. (2019)		✓		Min used A/C and delays	VNS, B&B	20	667	570.2 s

RN: route network; CN: connection network; TSN: time-space network; B&B: branch and bound; BD: Benders decomposition; B&P: branch and price; CG: column generation; DW: Dantzig-Wolfe decomposition; GA: genetic algorithm; LR: Lagrangian relaxation; RG: row generation; VNS: variable neighbourhood search; –: not reported.

The representative aircraft routing model of Liang and Chaovalitwongse (2013) is given in Table 8 and Constraints (12)–(15), generating a generic rotation in a weekly rotation-tour time-space network satisfying maintenance requirements rather than assigning individual aircraft to flights. The associated network is presented in Fig. 3. To be specific, the complete weekly network consists of seven three-day (equivalent to the maximum days between two maintenance instances MD) time-space subnetworks from Monday to Sunday. A feasible aircraft rotation can visit different subnetworks via maintenance arcs as indicated with red solid lines. Every maintenance arc starts from the end of the day n at a maintenance station m in the p th subnetwork and ends at the same station (start of day) in the $(n + 1) \bmod 7$ subnetwork:

$$(M3) \quad \min \sum_{p \in P} \sum_{h \in H_p} c_h y_{hp} \quad (12)$$

$$\sum_{p: f \in F_p} x_{fp} = 1 \quad \forall f \in F \quad (13)$$

$$\begin{aligned} & \sum_{f \in F} \sum_{p: f \in F_p} \alpha_{fpv}^+ x_{fp} + \sum_{m \in M} \sum_{n \in N} \beta_{mnpv}^+ z_{mpn} + \sum_{h \in H} \sum_{p: h \in H_p} \gamma_{hpv}^+ y_{hp} \\ & + w_{in} = w_{in} + \sum_{f \in F} \sum_{p: f \in F_p} \alpha_{fpv}^- x_{fp} + \\ & \sum_{m \in M} \sum_{n \in N} \sum_{p \in P} \beta_{mnpv}^- z_{mpn} + \sum_{h \in H} \sum_{p: h \in H_p} \gamma_{hpv}^- y_{hp} \quad \forall v \in V \quad (14) \end{aligned}$$

$$\sum_{f \in F_C} x_{fp} + \sum_{g \in G_C} z_{gpn} + \sum_{l \in L_C} w_l \leq K \quad (15)$$

$$\sum_{n \in \{1..MD\}} z_{mpd} \leq Q_{mp} \quad \forall m \in M, p \in P. \quad (16)$$

In this model, the objective function (12) aims to minimize the short connect penalty cost or negative through revenue. Constraints (13) require that every flight is covered once in the weekly rotation network. Constraints (14) are flow balance constraints to ensure that the inbound flight flow equals the outbound flow at each vertex. Constraints (15) are aircraft count constraints that depend on the selection of the count time. Lastly, Constraints (16) ensure that the number of aircraft being maintained at each station/day is not greater than the maintenance capacity.

4.2 Connection network-based models

Within connection network-based models, maintenance constraints are explicitly formulated with additional constraints and variables. In Clarke et al. (1997), a model that aims to find feasible maintenance rotations was developed to maximize the through value following the formulation of the asymmetric travelling salesman problem. While this publication failed to realize all maintenance requirements, Haouari et al. (2013) presented a compact non-linear model with direct expressions of maintenance requirements (max flying time, max number of takeoffs, and max number of days). They further applied reformulation and linearization techniques to overcome the difficulties in computing. By duplicating edges in the connection network to represent three cases, before maintenance, during maintenance, and after maintenance, Başdere and Bilge (2014) solved the operational ARP with remaining time consideration using branch and bound and compressed annealing. Noticing that past literature usually took some of the maintenance checks into consideration, causing varying maintenance demand, Safaei and Jardine (2018) presented a novel mathematical model based on the connection network to ensure that sufficient maintenance opportunities are offered. This procedure is done by dividing

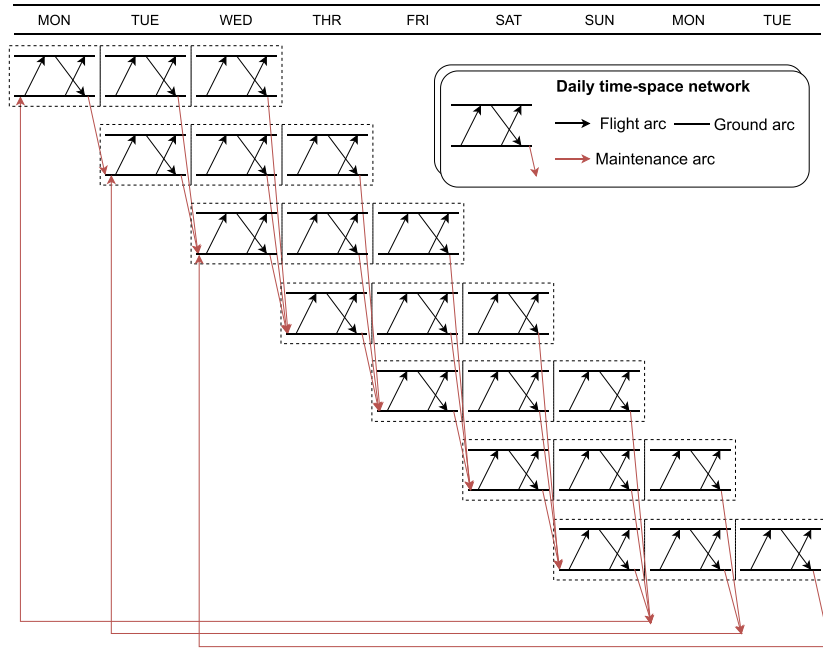


Figure 3 Weekly rotation-tour time-space network.

the planning horizon into a set time grid and posing additional constraints to count the total maintenance opportunities for an aircraft.

In this section, we present an operational model (given the remaining time of each aircraft before maintenance) of Başdere and Bilge (2014) given Table 9 and Constraints (17)–(26). It is worth noticing that this model tackles the aircraft routing problem with an operational planning horizon where aircraft initial positions are taken into consideration (referred to as the tail assignment problem):

$$(M4) \quad \min \sum_{a \in A_C} \left(R_a r_a - \sum_{f_1 \in F} t_{f_1} \left(\sum_{f_2 \in S(f_1)} x_{f_1 f_2 a} + \sum_{f_2 \in S'(f_1)} y_{f_1 f_2 a} \right) \right) \quad (17)$$

$$\sum_{f \in F(a)} (x_{ofa} + y_{ofa} + z_{ofa}) = 1 \quad \forall a \in A_C \quad (18)$$

$$\sum_{f \in F(a)} z_{ofa} = 1 \quad \forall a \in A_{C'} \quad (19)$$

$$\sum_{f \in F} z_{fta} = 1 \quad \forall a \in A \quad (20)$$

$$\begin{aligned} \sum_{f_2 \in P(f_1) \cup \{o\}} x_{f_2 f_1 a} &= \sum_{f_2 \in S(f_1)} x_{f_1 f_2 a} \\ &+ \sum_{f_2 \in S'(f_1)} y_{f_1 f_2 a} \quad \forall f_1 \in F, a \in A_C \quad (21) \end{aligned}$$

$$\sum_{f_2 \in P(f_1) \cup \{o\}} z_{f_2 f_1 a} = \sum_{f_2 \in S(f_1) \cup \{t\}} z_{f_1 f_2 a} \quad \forall f_1 \in F, a \in A_{C'} \quad (22)$$

$$\begin{aligned} \sum_{f_2 \in P(f_1) \cup \{o\}} y_{f_2 f_1 a} + \sum_{f_2 \in P(f_1) \cup \{o\}} z_{f_2 f_1 a} \\ = \sum_{f_2 \in S(f_1) \cup \{t\}} z_{f_1 f_2 a} \quad \forall f_1 \in F, a \in A_C \quad (23) \end{aligned}$$

$$\begin{aligned} \sum_{a \in A} \left(\sum_{f_2 \in P(f_1) \cup \{o\}} (x_{f_2 f_1 a} + z_{f_2 f_1 a}) + \sum_{f_2 \in P(f_1) \cup \{o\}} y_{f_1 f_2 a} \right) &= 1 \quad \forall f \in F \quad (24) \end{aligned}$$

$$\begin{aligned} \sum_{f_1 \in F} t_{f_1} \left(\sum_{f_2 \in S(f_1)} x_{f_1 f_2 a} + \sum_{f_2 \in S'(f_1)} y_{f_1 f_2 a} \right) &\leq R_a \quad \forall a \in A_C \quad (25) \end{aligned}$$

$$\begin{aligned} \sum_{f_1 \in F} \sum_{f_2 \in S(f_1) \cup \{t\}} t_{f_1} z_{f_1 f_2 a} &\leq R_a \sum_{f \in F} z_{ofa} \\ &+ L(1 - \sum_{f \in F} z_{ofa}) \quad \forall a \in A_C \quad (26) \end{aligned}$$

$$r_a = \sum_{f_1 \in F \cup \{o\}} \sum_{f_2 \in S(f_1)} y_{f_1 f_2 a} \quad \forall a \in A_{C'} \quad (27)$$

The objective function (17) is the minimization of unused flying time of the critical aircraft whose initial remaining time is less than the largest one in the connection network. Constraints (18) and (19) ensure that each aircraft in the fleet starts its route by flying one of the initial flights. The termination condition of each aircraft route is satisfied in Constraints (20). For a critical aircraft, flow balance constraints (21) indicate that, if the aircraft covers a flight before maintenance, the next executed flight must be either before maintenance again or right after maintenance. For non-critical aircraft, Constraints (22) restrict the routes within non-maintenance connections. Constraints (23) model the flow balance after aircraft maintenance. Flight cover constraints (24) ensure that every flight is served by one aircraft. Constraints (25) restrict the legal flying hours before maintenance and Constraints (26) are restrictive for critical aircraft that do not receive maintenance. Finally, Constraints (27) link variables r and y . The overall model formulation thus ensures the adherence to maintenance rules during the planning horizon.

Table 8. Additional notation for the time-space network-based aircraft routing model M3.

Sets	
N_d	Set of days of the planning, $p \in N_d$.
S_p	n -day time-space network starts on day $p \in P = \{1, 2, \dots, 7\}$.
F_p	Set of flight edges in the S_p network.
H	Set of connected edges in the weekly rotation-tour network (WRTN), $h \in H$.
H_p	Set of connected edges in the S_p network.
M	Set of maintenance stations, $m \in M$.
G	Set of maintenance arcs in WRTN, $g \in G$.
V	Set of vertices in WRTN, $v \in V$.
F_C	Set of flight edges passing the count time in WRTN.
H_C	Set of connected arcs passing the count time in WRTN.
G_O	Set of maintenance edges passing the count time in WRTN.
L	Set of ground edges in WRTN, $l \in L$.
l_v^+	Ground edge before vertex v .
l_v^-	Ground edge after vertex v .
$Q_{m,p}$	Maintenance capacity at station m on day p .
c_h	Cost of connected arc $h \in H$.
MD	Maximum days between two consecutive maintenance instances.
Parameters	
α_{fpu}^+	1 if flight f in S_p network starts at vertex u , 0 otherwise.
α_{fpu}^-	1 if flight f in S_p network ends at vertex u , 0 otherwise.
γ_{fpu}^+	1 if edge h in S_p network starts at vertex u , 0 otherwise.
γ_{fpu}^-	1 if edge h in S_p network ends at vertex u , 0 otherwise.
β_{mpnu}^+	1 if maintenance edge (station m , day n) in S_p network starts at vertex u , 0 otherwise.
β_{mpnu}^-	1 if maintenance edge (station m , day n) in S_p network ends at vertex u , 0 otherwise.
K	Size of the aircraft fleet.
Decision variables	
$x_{f,p}$	1 if flight f is flown in the S_p network, 0 otherwise.
z_{mpn}	Number of aircraft on maintenance edge at station m , S_p network, and day n .
w_l	Number of aircraft on ground edge l .

4.3 Route network-based models

The route network is composed of aircraft routes which are sequences of connected flights that are maintenance feasible. Therefore, complex maintenance constraints can be simplified and embedded in the variable definition. First proposed by [Barnhart et al. \(1998\)](#), the developed model consists of flight strings that start and end at maintenance stations with additional time left for maintenance. Hence, a flight string is a special case of the maintenance feasible routes mentioned in this part. To generate these flight strings, the branch-and-price algorithm was applied in this context with the follow-on branching scheme. In [Gopalan and Talluri \(1998\)](#); [Talluri \(1998\)](#), they chose to generate lines of flying (LOF) which are connected flights within one day through finding k -day maintenance Euler tours with a designed polynomial time algorithm. As a consequence, the problem can only be modelled without an objective function that limits its application. Focusing on solving the operational problem rather than the tactical problem, [Sarac et al. \(2006\)](#) considered the legal

Table 9. Additional notation for the connection network-based aircraft routing model M4.

Sets	
A_C	Set of critical aircraft.
$A_{C'}$	Set of non-critical aircraft.
$F(a)$	Set of flights that aircraft a can fly.
$S(f)/P(f)$	Set of flights immediately succeeding/preceding flight f .
$S'(f)/P'(f)$	Set of flights immediately succeeding/preceding flight f and allowing maintenance.
$\{o, t\}$	Artificial source and sink node of flight network.
Parameters	
t_f	Duration of flight f .
R_a	Remaining time of aircraft a .
L	Length of the longest route on the connection network in terms of accumulated flight time.
Decision variables	
$x_{f_1f_2a}$	1 if aircraft a flies flights f_1 and f_2 consecutively before replenishment, 0 otherwise.
$y_{f_1f_2a}$	1 if aircraft a flies flights f_1 and f_2 consecutively and undergoes maintenance in between, 0 otherwise.
$z_{f_1f_2a}$	1 if aircraft a flies flights f_1 and f_2 consecutively either after a replenishment or in a route with no replenishment, 0 otherwise.
r_a	1 if aircraft a undergoes a maintenance operation, 0 otherwise.

remaining hours of different aircraft and resource constraints at maintenance stations. Due to the introduction of resource constraints, they developed two additional branching schemes as a supplementation to the follow-on branching scheme. Lastly, [Maher et al. \(2018\)](#) developed a daily operational aircraft routing problem (tail assignment problem) that implicitly satisfies day-two and day-three maintenance requirements using look-ahead constraints which differs from the aforementioned implementations concerning maintenance constraints.

Because the computation of propagated delays is non-linear, the route network-based model is also widely applied to robust operation planning and there are two approaches (*stability, recoverability*) in achieving robustness. Regarding the stability-based approach where generated schedules are less likely to be impacted by delay disruptions, [Lan et al. \(2006\)](#) proposed two models involving reducing propagated delays and minimizing passenger missed connections by re-timing. As the propagated delays can be modelled in the objective function, this formulation eases the computation efforts in incorporating robustness. Similarly, a robust weekly ARP was solved in the work of [Liang et al. \(2015\)](#) to minimize the expected propagated delays. By duplicating daily time-space subnetworks and generating lines of flight, the model is compact with fewer decision variables in contrast to other route network-based models. Column generation is applied as well to picking promising aircraft routes from the complete network. One shortcoming of this model is that only the maximum accumulated calendar days can be modelled as the maintenance requirement. Focusing on delay correlation, the model of [Yan and Kung \(2018\)](#) utilized robust optimization to protect the generated flight schedule from being disturbed by propagated delays defined by an uncertainty set. To ease the computation complexity, the column generation and cutting plane method was applied for robust solutions. From the recoverability side where flight schedules are recoverable

Table 10. Additional notation for the route network-based aircraft routing model M5.

Sets	
S	Set of flight strings, $s \in S$.
S_{f-}	Set of augmented strings ending with flight f .
S_{f+}	Set of augmented strings beginning with flight f .
G	Set of ground edges, $j \in G$.
Parameters	
a_{fs}	1 if flight f is contained in string s .
r_s^k	Number of times string s of fleet k crosses the count time.
p_j^k	Number of times ground edge j of fleet k crosses the count time.
$e_{f,a}/e_{f,d}$	Event number corresponding to the arrival/departure of flight f .
$e_{f,a}^-/e_{f,d}^-$ $e_{f,a}^+/e_{f,d}^+$	Event preceding the arrival/departure of flight f . event after the arrival/departure of flight f .
c_s^k	Cost of flying augmented string s with fleet k .
Decision variables	
x_s^k	1 if string s is flown by fleet k in the solution, 0 otherwise.
$y_{e_1 e_2}^k$	Number of aircraft of fleet k on ground between two events e_1, e_2 .

within limited efforts once being disrupted, Froyland et al. (2014) integrated the operational aircraft routing problem with a recovery problem with over 100 disruption scenarios. To solve this stochastic programming model, improved Benders decomposition and column generation algorithms are jointly used to derive optimal robust solutions.

The mathematical formulation of the classic model from Barnhart et al. (1998) is presented in Table 10 and Constraints (28)–(32). Although fleeting decisions are also incorporated, the model can be adapted to derive routes for aircraft:

$$(M5) \quad \min \sum_{k \in K} \sum_{s \in S} c_s^k x_s^k \quad (28)$$

$$\sum_{k \in K} \sum_{s \in S} a_{fs} x_s^k = 1 \quad \forall f \in F \quad (29)$$

$$\sum_{s \in S_f} x_s^k - y_{e_{f,d}^-, e_{f,d}^+}^k + y_{e_{f,d}^-, e_{f,d}^+}^k = 0 \quad \forall f \in F, k \in K \quad (30)$$

$$- \sum_{s \in S_f} x_s^k - y_{e_{f,a}^-, e_{f,a}^+}^k + y_{e_{f,a}^-, e_{f,a}^+}^k = 0 \quad \forall f \in F, k \in K \quad (31)$$

$$\sum_{s \in S} r_s^k x_s^k + \sum_{j \in G} p_j^k y_j^k \leq N_k \quad \forall k \in K. \quad (32)$$

In this model, Constraints (29) are set partition constraints that ensure that every flight is covered in one string. The flow balance constraints given in (30) and (31) are adopted to establish a repeated schedule. The number of available aircraft is restricted by Constraint (32) and in our weekly planning horizon, $r_s, p_j \leq 1$. The model is solved by branch and price and the pricing problem (resource constraint shortest path problem) is solved by a multi-label shortest path to ensure that the maximum flying time and the maximum flying day restrictions are satisfied. As the number of feasible flight strings is too large to be enumerated explicitly, the branch-and-price algorithm is used to solve the model. In particular, column generation is first applied to solve the linear relaxation model of M5 where integer variables x_s are relaxed.

In summary, research in ARP has paid a great deal of attention to operational planning aspects with very realistic maintenance constraints. Additionally, routing-related robustness is being broadly studied from stability and recoverability aspects using either historical statistics or domain-specific knowledge. Concerning future research insights, a wide range of uncertainty in terms of flight time and maintenance capacity, along with more realistic and general operational considerations like more accurate fuel consumption cost approximation, is expected to gradually gain popularity. Additionally, the objective function of ARP can vary from minimizing operation costs and improving utilization rate to decreasing propagated delays. Consequently, the multi-objective optimization model with the corresponding solution algorithm is another promising point to cater to the different requirements of airlines.

5. Crew scheduling problem

After the aircraft routing process, crew scheduling is carried out to allocate air cabin/cockpit crew to a flight; this is commonly divided into two subproblems: the *crew pairing problem* (CPP) and the *crew rostering problem* (CRP) due to high complexity. In CPP, a set of crew pairings (consisting of several daily duties) composed of a sequence of flights is generated with minimum pairing cost. Thereafter, monthly schedules for each crew member are generated by CRP with respect to crew rules and preferences over vocation, training, and so on. The involved literature is summarized in Table 11 where the Horizon column implies the planning horizon of the crew scheduling problem.

Concerning existing review papers on the crew scheduling problem, Kasirzadeh et al. (2017) studied the existing literature from the aspects of CPP, CRP, and the integrated crew scheduling problem. Additionally, a model for the sequential personalized crew assignment problem is proposed. Their suggestions for future works in terms of CSP include sophisticated types of crew preferences, robust schedules against the uncertainty of perturbations, and simultaneous construction of schedules for cabin and cockpit crews. Further, in the survey paper of Deveci and Demirel (2018), model formulation types and different solution algorithms were summarized. While robustness is also mentioned as one future research direction, they also suggest casting efforts toward fairness in crew work distribution and the tiredness of the team in terms of sleep loss and irregular sleep.

5.1 Crew pairing models

Because the pairing cost is computed as the summation of duty cost, which depends on a maximum of flying hours, elapsed time, and minimum guaranteed time, CRP is usually formulated as a set partition/covering problem to facilitate the cost computation and crew rule checking. Besides, CPP can be classified into three types: daily, weekly, and dated (monthly) problems according to different planning horizons; the daily problem is extensively studied in the literature.

In the basic crew pairing model, Lavoie et al. (1988) model pairings as decision variables and set partition/covering constraints are set up to ensure that every flight is covered by at least one pairing. However, the cost structure considered here is a linear approximation to the real cost, which causes inaccuracy. The set partition model can be efficiently solved by the branch-and-price algorithm presented by Hoffman and Padberg (1993) to solve large-scale problems. Based on this model, Barnhart et al. (1995) further incorporated deadhead decisions to improve crew utilization. The candidate deadhead flights are selected

Table 11. Summary of the literature on the crew scheduling problem.

Literature	CPP	CRP	Horizon	Formulation	Objective	Method	Crew(bases)	Flight	Time
Lavoie et al. (1988)	✓		–	SC	Min pairing cost	CG	–	1,113	1,250 s
Hoffman and Padberg (1993)	✓		–	SP	Min pairing cost	B&P	–	–	1,410.6 s
Barnhart et al. (1995)	✓		–	SP	Min pairing and deadhead cost	CG	2	833	–
Dawid et al. (2001)		✓	Dated	SP	Max crew utility value	B&B	1,300	–	605.04 s
Makri and Klabjan (2004)	✓		Dated	SP	Min pairing cost	CG	6	337	1,860 s
Schaefer et al. (2005)	✓		Daily	SP	Min expected pairing cost	Local search	–	342	–
Guo et al. (2006)	✓	✓	Weekly	NF	Min overnight and compensation cost	Assignment heuristic	188	1,977	401 s
Yen and Birge (2006)	✓		Daily	SP	Min pairing cost	B&P	–	79	–
Medard and Sawhney (2007)	✓	✓	Dated	SC	Min pairing and rostering cost	CG	439	–	492 s
Souai and Teghem (2009)	✓	✓	Daily	SP	Min total cost and deviation	GA, local search	68	1,872	3,640 s
Maenhout and Vanhoucke (2010)		✓	Dated	SP	Min total penalty cost	Scatter search heuristic	–	–	127.21 s
Saddoune et al. (2012)	✓	✓	Dated	SP	Min schedule and pilot penalty cost	CG, constraint aggregation	54	7,527	3,451.68 s
Muter et al. (2013)	✓		Weekly	SC	Min pairing and deadhead cost	CG	1	490	28.31 s
Doi et al. (2018)		✓	Dated	SP	Min deviation from working time	Partial optimization metaheuristic under special intensification conditions	289	–	7,420 s
Antunes et al. (2019)	✓		Daily	SP	Min pairing and delay cost	CG	–	94	14,400 s
Haouari et al. (2019)	✓		Daily	NF	Min layover time	Reformulation linearization, B&B	172	336	7,705.18 s
Wen et al. (2020)	✓		Daily	SC	Min pairing and robustness cost	CG	–	98	–
Zeighami et al. (2020)	✓	✓	Monthly	SP	Max crew requests, min pairing cost, min dissimilarity of pilot and co-pilot	CG, ALD	3	5,613	1,943.52 s
Tahir et al. (2021)	✓		Weekly	SP	Min work time, deadhead, short connection and short rest	Integral column generation, convolutional neural network	3	1,694	696 s

CPP: crew pairing problem; CRP: crew rostering problem; NF: network flow model; SC: set covering model; SP: set partition model; B&B: branch and bound; B&P: branch and price; CG: column generation; GA: genetic algorithm; ALD: alternative Lagrangian decomposition; –: not reported.

through a successive elimination process and a great amount of crew cost is reduced owing to this extension. Focusing on the column generation implementation for CPP, Makri and Klabjan (2004) developed several pruning methods regarding solving the pricing subproblem. In particular, the pruning rules use shortest-path algorithms to get bounds for pruning unpromising labels. Apart from the set partition model, one novel compact model was presented by Haouari et al. (2019) using non-linear constraints. As a result, reformulation and linearization techniques are applied to the original model for better computation performance and valid inequalities are adopted to tighten the model. One advantage of the presented model is the applicability to general commercial solvers instead of sophisticated branch-and-price algorithms. More recently, there has been growing interest in

combining machine learning techniques with the crew scheduling problem. One representative work by Tahir et al. (2021) utilizes the deep convolutional neural network to predict flight connection probabilities. These values are embedded in a partial pricing scheme to reduce the computation time.

Another important extension to CRP is robustness and uncertainty modelling. In Schaefer et al. (2005), instead of using planned cost to evaluate the quality of a crew schedule, expected operation cost was adopted in this study with a push-back aircraft recovery simulation tool under uncertain disruptions. Yen and Birge (2006), on the other hand, noticed the inefficiency stemmed from crew changing aircraft where the minimum sit time is requested and delay caused by aircraft routing may be exacerbated due to inappropriate assignment of these connections under multiple

Table 12. Additional notation for the crew pairing model M6.

Sets	
Q	Set of indices for the side constraints.
Parameters	
c_p	Cost of pairing p .
a_{fp}	Number of times that flight $f \in F$ is actively covered in pairing $p \in P$.
b_{qp}	Contribution of pairing $p \in P$ to side constraint $q \in Q$.
e_q	Right-hand side of side constraint $q \in Q$.
Decision variables	
y_p	1 if pairing p is selected, 0 otherwise.

disruption scenarios. To solve this stochastic programming model, specific branching rules were designed. Another model variant presented by Antunes et al. (2019) utilized robust optimization to generate robust crew pairing solutions where a disruption distribution model is not necessary. Both the delay propagation through crew connections and the crew salary scheme were incorporated. Finally, flight time variability was taken into account in the work by Wen et al. (2020) with novel robustness-related costs being established.

CPP is usually formulated as a set partition model problem, and we present a basic crew pairing model proposed by Saddoune et al. (2013) with Constraints (33)–(35) and the definitions of variables and other data in Table 12:

$$(M6) \quad \min \sum_{p \in P} c_p y_p \quad (33)$$

$$\sum_{p \in P} a_{fp} y_p = 1 \quad \forall f \in F \quad (34)$$

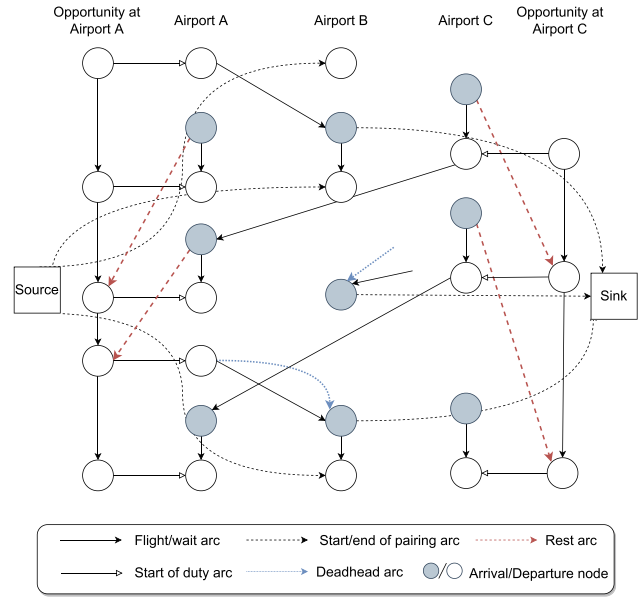
$$\sum_{p \in P} b_{qp} y_p = e_q \quad \forall q \in Q. \quad (35)$$

The objective function (33) minimizes the total cost of the selected pairings. Constraints (34) ensure that exactly one active crew is assigned to every flight. Constraints (35) include supplementary restrictions such as the number of available crews at each base.

The complexity of the model lies in the construction of feasible pairings from a crew connection network shown in Fig. 4 where common pairing and duty rules (e.g. minimum rest time, the maximum amount of flying time per duty) have to be considered. The model then is solved by a rolling horizon approach which is composed of iterative time slices and the column generation method.

5.2 Crew rostering models

In contrast to CPP, CRP focuses more on practical considerations where both satisfaction and fairness should be taken into consideration. While the basic crew rostering model only includes crew count constraints and minimum required crew constraints, a group of researchers have extended this problem from different viewpoints. For example, Dawid et al. (2001) consider an enhanced rostering model where crew members are allowed to be downgraded to meet demand and thus improve the flexibility level. In Doi et al. (2018), fair working time of crew members was achieved by penalizing deviation from standard assigned values. The solution algorithm of this model is a metaheuristic and the original model is decomposed into a subproblem and a

**Figure 4** Connection network for the crew pairing model M6, where Airport B is the base station.

master problem, which is further reformulated as a generalized set partition model.

One recent trend in CSP-related studies is to integrate CRP with CPP to yield global optimal crew schedules. A first attempt by Guo et al. (2006) incorporated preassigned activities into a crew pairing problem, which facilitates the solving of CRP and a direct reduction in the total crew cost. To further tackle large-scale real-life problems, Medard and Sawhney (2007) integrated the pairing and rostering models with a restriction on the recovery time period to control the model size. The problem was solved by a simple tree search and a more sophisticated column generation method. Other solution techniques are also applied by researchers for greater computation efficiency, e.g. genetic algorithms (Souai and Teghem 2009) and constraint aggregation (Saddoune et al. 2012). More recently, Zeighami et al. (2020) introduced an integrated model to generate schedules for pilots and co-pilots simultaneously. Both delay propagation and dissimilarity between pilot schedules and co-pilot schedules were optimized. A hybrid solution algorithm that combines alternative Lagrangian decomposition, column generation, and dynamic constraint aggregation was developed.

The integrated crew pairing and personalized crew assignment model from Zeighami et al. (2020) is illustrated here with additional parameters, variables, and formulation as shown in Table 13 and Constraints (36)–(43):

$$(M7) \quad \min \sum_{l \in L} \sum_{s \in S_l} c_s x_s + \sum_{l \in L} \sum_{v \in V_l} c_v r_v + \sum_{o \in O} \sum_{s \in S_o} c_s y_s + \sum_{o \in O} \sum_{v \in V_o} c_v z_v + \sum_{(f_1, f_2) \in C} R_{f_1 f_2} u_{f_1 f_2} \quad (36)$$

$$\sum_{l \in L} \sum_{s \in S_l} e_f^s x_s = 1 \quad \forall f \in F \quad (37)$$

$$\sum_{s \in S_l} b_v^s x_s + r_v = 1 \quad \forall l \in L, \forall v \in V_l \quad (38)$$

Table 13. Additional notation for the integrated crew pairing and personalized crew assignment model M7.

Sets	
L	Set of pilots, $l \in L$.
O	Set of co-pilots, $o \in O$.
V_l/V_o	Set of vocation requests for pilot l /co-pilot o , $v \in V_l/V_o$.
S_l/S_o	Set of feasible schedules for pilot l /co-pilot o .
P_s	Set of all pairings covered by schedule s .
C	Set of flight connections, $(f_1, f_2) \in C$.
Parameters	
e_f^s	1 if flight f is covered by schedule s , 0 otherwise.
b_v^s	1 if vocation request v is satisfied by schedule s .
c_p	Cost of pairing p .
n_s	Number of preferred flights in schedule s .
B	Bonus cost for covering each preferred flight.
c_s	Cost of schedule s , $c_s = \sum_{p \in P_s} c_p + B n_s$.
$a_{f_1 f_2}^s$	1 if flight connection between f_1 and f_2 is covered by schedule s .
$R_{f_1 f_2}$	Penalty cost for $u_{f_1 f_2}$.
Decision variables	
x_s	1 if schedule $s \in S_l$ is assigned to pilot l , 0 otherwise.
y_s	1 if schedule $s \in S_o$ is allocated to co-pilot o , 0 otherwise.
r_v	1 if vocation request $v \in V_l$ is unsatisfied for pilot l , 0 otherwise.
z_v	1 if vocation request $v \in V_o$ is unsatisfied for co-pilot o , 0 otherwise.
$u_{f_1 f_2}$	1 if flight f_1 and f_2 is covered by either pilot or co-pilot problem, 0 otherwise.

$$\sum_{s \in S_l} x_s \leq 1 \quad \forall l \in L \quad (39)$$

$$\sum_{l \in L} \sum_{s \in S_l} a_{f_1 f_2}^s x_s - \sum_{o \in O} \sum_{s \in S_o} a_{f_1 f_2}^s y_s \leq u_{f_1 f_2} \quad \forall (f_1, f_2) \in C \quad (40)$$

$$\sum_{o \in O} \sum_{s \in S_o} e_f^s y_s = 1 \quad \forall f \in F \quad (41)$$

$$\sum_{s \in S_o} b_v^s y_s + z_v = 1 \quad \forall o \in O, \forall v \in V_o \quad (42)$$

$$\sum_{s \in S_o} y_s \leq 1 \quad \forall o \in O \quad (43)$$

The objective function (36) aims to reach a trade-off between maximizing the number of satisfied vocation requests along with preferred flights and minimizing the total pairing cost with the dissimilarity between pilot and co-pilot pairings. For pilot-related restrictions, set partition constraints (37) ensure that every flight is covered by one pilot. Constraints (38) capture the allocation status of pilot vocation requests. At most one schedule can be distributed to one pilot as specified by Constraints (39). Similar restrictions are imposed on co-pilots in Constraints (41)–(43). Lastly, linking Constraints (40) capture the inconsistency between pilot and co-pilot schedules.

To conclude, the general trend in crew scheduling-related research is the integration of crew pairing and crew assignment (rostering) along with crew schedule robustness against delays. Whilst extensive publications have tackled these two points, there remain several directions for future work. Firstly, hybrid solution algorithms that combine typical decomposition algorithms (e.g. column generation) with fast heuristics and accelerating skills

ought to be developed for better computational performance. Secondly, new compact mathematical formulations as in Haouari et al. (2019) should be further developed to provide efficient and easily implementable alternatives. Finally, more delicate robustness characteristics like delay correlations and crew availability uncertainty are expected to offer more reliable schedules.

6. Integrated problems

The widely used sequential scheduling models reduce the problem complexity significantly with sacrifices regarding global optimality and feasibility, as the solution from one subproblem may not necessarily be the optimal or even a feasible solution for its succeeding subproblem. Hence, many enhancements have been implemented in the recent decade in terms of integrating two or more subproblems. The integrated scheduling publications involved are summarized in Table 14 and Fig. 5 illustrates the correlation and basic integration concept between two subproblems.

6.1 Integrated schedule design and fleet assignment problem

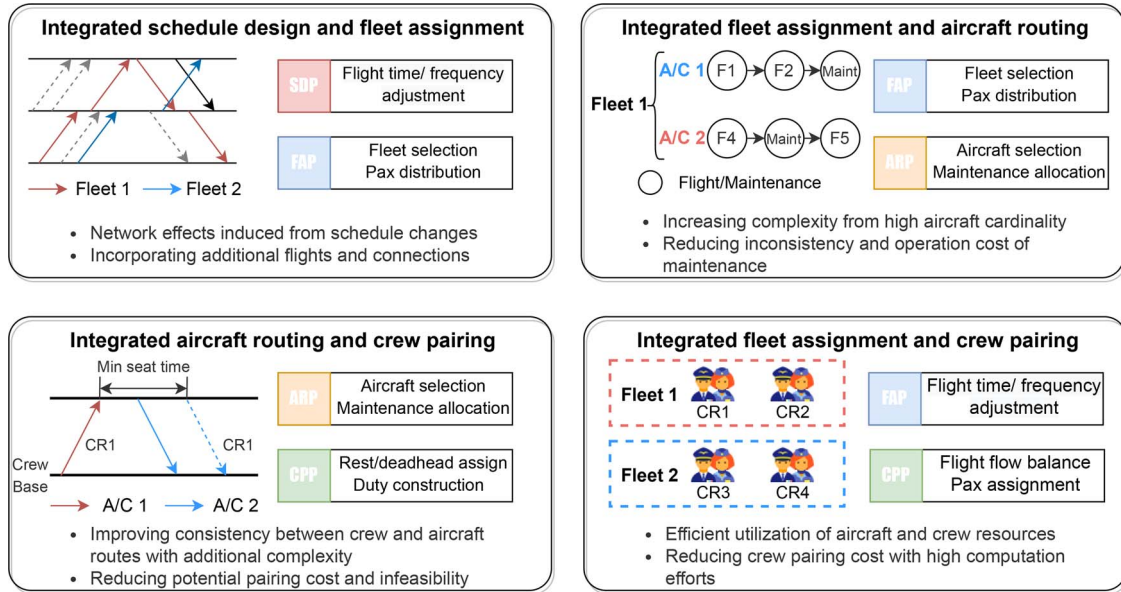
Schedule design decisions can be combined with FAP to generate higher revenue through demand–supply interaction. In general, schedule design approaches can be classified into *incremental design* and *comprehensive design* depending on whether the schedule is built from an existing schedule, and most current works belong to the former class using re-timing and optional flight decisions. Lohatepanont and Barnhart (2004) presented two models that simultaneously optimize flight selection and fleet assignment decisions. Demand correction was considered regarding flight cancellation and passenger recapture rate. A heuristic branch-and-bound (B&B) algorithm that combined column generation and row generation was developed to solve instances from a major US airline. Jiang and Barnhart (2009) proposed a schedule re-optimization model that supplemented re-fleeting with re-timing as a dynamic scheduling mechanism to cope with demand fluctuation. B&B with a Type I special ordered set branching scheme was adopted as the solution approach. Focusing on a subsidized air transportation network, Pita et al. (2014) proposed a socially oriented flight scheduling and fleet assignment model for reducing total social cost through network configuration. Cadarso and Celis (2017) presented an integrated model with the optimization of schedule design, fleet assignment, and passenger-related factors. Using data from a Spanish airline, an improved Benders decomposition solution algorithm was developed and the benefits of integrated approaches were shown. Wei et al. (2020) proposed a mixed integer model for comprehensive timetabling and FAP with consideration of endogenous passenger choices. The highly intractable model is solved with a multiphase solution approach and corresponding rule-based acceleration strategies. Based on the network of Alaska Airlines, the advantages of the comprehensive scheduling approach are illustrated in contrast to the incremental approach. A novel flight scheduling and fleet assignment model with endogenized demand generation and allocation was developed in Birolini et al. (2021) to capture the demand–supply interaction. A real-world case study further validated its efficiency in increasing the operating margin by 6.9 per cent.

In this section, the approximate schedule design and fleet assignment model of Lohatepanont and Barnhart (2004) is presented to demonstrate the integration of schedule design and fleet assignment. Given the additional notation in Table 15, the

Table 14. Summary of the literature for integrated airline scheduling.

Literature	SDP	FAP	ARP	CSP	Flights	Aircraft	Model	Method	Time
Lohatepanont and Barnhart (2004)	✓	✓			848	166	MIP	CG	45,720 s
Sandhu and Klabjan (2007)		✓		✓	942	–	MIP	LR, BD	29 h
Jiang and Barnhart (2009)	✓	✓			1,000	–	MIP	B&B	20 h
Papadakis (2009)		✓	✓	✓	705	167	MIP	BD, CG	27.8 h
Sherali et al. (2010)	✓	✓			1,476	–	MINP	Valid inequalities, BD	2,988 s
Haouari et al. (2011)		✓	✓		1,050	34	MIP	BD	58 s
Jiang and Barnhart (2013)	✓	✓			1,000	–	MIP	B&B	36,000 s
Sherali et al. (2013)	✓	✓	✓		592	–	MINP	BD	43,200 s
Pita et al. (2014)	✓	✓			325	24	MIP	B&B	72,360 s
Salazar-González (2014)		✓	✓	✓	156	16	MIP	B&B, four-step heuristic	254.1 s
Ruther et al. (2016)			✓	✓	1,130	50	MIP	CG, Diving	7,815 s
Cacchiani and Salazar-González (2017)		✓	✓	✓	172	17	MIP	B&P	5 h
Cadarso and Celis (2017)	✓	✓			–	47	MIP	BD	3841 s
Faust et al. (2017)	✓	✓	✓		–	15	MIP	B&P	143 s
Shao et al. (2017)		✓	✓	✓	676	192	MIP	BD, CG	594.8 min
Kenan et al. (2018)	✓	✓	✓		228	59	MIP	CG	6,854 s
Wei et al. (2020)	✓	✓			390	–	MIP	B&B, rules	2 h
Parmentier and Meunier (2020)			✓	✓	1,766	59	MIP	CG, B&B	23,671 s
Xu et al. (2021)	✓	✓	✓		1,607	51	MINP	Variable neighbourhood search, CG	3,460 s
Birolini et al. (2021)	✓	✓			–	–	MINP	B&B	6,459 s
Ahmed et al. (2022)		✓	✓	✓	646	202	MIP	Proximity search, B&B	18,000 s
Yan and Ghate (2022)	✓	✓			179	34	MIP	LR	–

MIP: mixed integer programming; MINP: mixed integer non-linear programming; CG: column generation; BD: Benders decomposition; B&B: branch and bound; B&P: branch and price; LR: Lagrangian relaxation; –: not reported.

**Figure 5** Integration scheme for airline schedule subproblems.

mathematical formulation is shown as follows:

$$(M8) \quad \min -R + \sum_{f \in F} \sum_{k \in K} c_{k,f} x_{k,f} + \sum_{i_1 \in I} \sum_{i_2 \in I} (fare_{i_1} - b_{i_1}^{i_2} fare_{i_2}) t_{i_1}^{i_2} \quad (44)$$

$$\text{s.t.} \quad \sum_{k \in K} x_{k,f} = 1 \quad \forall f \in F^M \quad (45)$$

$$\sum_{k \in K} x_{k,f} \leq 1 \quad \forall f \in F^O \quad (46)$$

$$y_{k,ap,t^-} + \sum_{f \in I_{k,ap,t}} x_{k,f} - y_{k,ap,t^+} \quad (47)$$

$$- \sum_{f \in O(k,ap,t)} x_{k,t} = 0 \quad \forall \{k, ap, t\} \in V \quad (48)$$

$$\sum_{ap \in AP} y_{k,ap,t_{\cap}} + \sum_{f \in CF(k)} x_{k,f} \leq N_k \quad \forall k \in K \quad (49)$$

$$\sum_{k \in K} S_k x_{k,f} + \sum_{i_1 \in I} \sum_{i_2 \in I} \delta_f^{i_1 i_2} t_{i_1}^{i_2} - \sum_{i_1 \in I} \sum_{i_2 \in I} \delta_f^{i_2 i_1} t_{i_1}^{i_2} \geq \sum_{i \in I} \delta_f^i D_i \quad f \in F \quad (50)$$

$$\sum_{i_2 \in I} t_{i_1}^{i_2} \leq D_{i_1} \quad \forall i_1 \in I. \quad (51)$$

Table 15. Additional notation for the integrated schedule design and fleet assignment model M8.

Sets	
F^M	Set of mandatory flights.
F^O	Set of optional flights.
T	Sorted set of all event times in increasing order, $t_j \in T$.
V	Set of vertices in the time-space network, $\{k, ap, t\} \in V$.
$I(k, ap, t)$	Set of inbound flights to vertex $\{k, ap, t\}$.
$O(k, ap, t)$	Set of outbound flights to vertex $\{k, ap, t\}$.
$CF(k)$	Set of flights that pass the count time when flown by fleet k .
Parameters	
R	Initial unconstrained revenue.
$b_{i_1}^{i_2}$	Recapture rate from itinerary i_1 to i_2 .
δ_f^i	1 if itinerary i includes flight f , 0 otherwise.
$C_{k,f}$	Cost of operating flight f with fleet k .
Decision variables	
$x_{k,f}$	1 if flight f is assigned to fleet k , 0 otherwise.
$t_{i_1}^{i_2}$	Number of passengers requesting itinerary i_1 but who are redirected to itinerary i_2 .
$y_{k,0,t_j^+}$	Number of aircraft in fleet k on the ground at airport immediately after t_j .
$y_{k,0,t_j^-}$	Number of aircraft in fleet k on the ground at airport immediately before t_j .

Similar to model M2, the integrated model relies on the time-space network and separates the executed flights into two disjoint sets as depicted in Constraints (35). Consequently, the isolation preprocessing techniques cannot be applied with the introduction of optional flights, which produces a larger model and higher complexity.

6.2 Integrated schedule design, fleet assignment, and aircraft routing problems

Models that integrate these three subproblems can be characterized by a high intrinsic computational complexity. Sherali et al. (2013) incorporated aircraft routing into their earlier integrated model (Sherali et al. 2010), taking into account flight re-timing and schedule balance. Linearization techniques and Benders decomposition were developed and the benefits of considering several subproblems simultaneously were demonstrated. Faust et al. (2017) presented an integrated schedule design and aircraft routing model for a point-to-point airline with a homogeneous fleet. Time-of-day specific demand modelling and non-linear revenue were adopted. B&P and two heuristics based on a subset of generated columns were developed as the solution methodologies. Kenan et al. (2018) integrated schedule design, fleet assignment, and aircraft routing to formulate a realistic model. Demand, delays, and deadhead flights were considered while the maintenance requirement was implicitly ensured. Column generation was used to obtain optimal solutions from a reformulated model. In the work of Xu et al. (2021), a robust scheduling model was presented to minimize the operation cost and expected delay. A hybrid solution algorithm that combines column generation and variable neighbourhood search methods is adopted to derive high-quality solutions within reasonable computation time.

In Table 16 and Constraints (52)–(66), we present the integrated model of Sherali et al. (2013):

$$(M9) \quad \max \sum_{i_1 \in I} \sum_{i_2 \in I_{i_1}} \text{fare}_{i_2} \pi_{i_1 i_2} - \sum_{f \in F} \sum_{a \in A} C_{fa} x_{af} \quad (52)$$

$$\sum_{a \in A(f)} x_{af} = 1 \quad \forall f \in F^M \quad (53)$$

$$\sum_{a \in A(f)} x_{af} \leq 1 \quad \forall f \in F^O \quad (54)$$

$$\sum_{f \in F(a): (ap, f) \in E} y_{a, ap+n, f} \leq 1 \quad \forall a \in A^{ap}, ap \in AP \quad (55)$$

$$\sum_{a \in A} \sum_{f: (n+ap, f_1) \in A} y_{a, ap, f} +$$

$$\sum_{f: (f, ap) \in C} \sum_{a \in A^{ap} \setminus A(f)} y_{a, ap, f} +$$

$$\sum_{f: (ap, f) \in E} \sum_{a \in A \setminus A(f)} y_{a, f, ap} = 0 \quad \forall ap \in AP \quad (56)$$

$$\sum_{a \in A^{ap}} \sum_{f \in F(a): (ap+n, f) \in E} y_{a, ap, f} =$$

$$\sum_{a \in A^{ap}} \sum_{f \in F(a): (f, ap) \in C} y_{a, f, ap} \quad \forall ap \in AP, \forall k \in K \quad (57)$$

$$\sum_{f_2: (f_1, f_2) \in A, f_2 \in F(a)} y_{af_1 f_2} = x_{af_1} \quad \forall f_1 \in F, a \in F(a) \quad (58)$$

$$\sum_{f_1: (f_1, f_2) \in A, f_1 \in F(a)} y_{af_1 f_2} = x_{af_2} \quad \forall f_2 \in F, a \in F(a) \quad (59)$$

$$z_i - \sum_{a \in A} x_{af} \leq 0 \quad \forall i \in I^O, f \in F^O(i) \quad (60)$$

$$z_i - \sum_{f \in F^O(i)} \sum_{a \in A} x_{af} \geq 1 - |F^O(i)| \quad \forall i \in I^O \quad (61)$$

$$\sum_{a \in A} x_{af} \leq \sum_{i \in I^O(f)} z_i \quad \forall f \in F^O \quad (62)$$

$$\sum_{i_1 \in I} \sum_{i_2 \in I_{i_1}} \pi_{i_1 i_2} \leq \sum_{a \in A} \text{Cap}_a x_{af} \quad \forall f \in F \quad (63)$$

$$\sum_{i_2 \in I_{i_1}} \pi_{i_1 i_2} \leq D_{i_1} \quad \forall i_1 \in I \quad (64)$$

$$\pi_{ii} \leq D_i z_i \quad \forall i \in I^O \quad (65)$$

$$\pi_{i_1 i_2} \leq \beta_{i_1 i_2} (D_{i_1} - \pi_{i_1 i_1}) z_{i_2} \quad \forall i_1 \in I, i_2 \in I_{i_1} \quad (66)$$

The objective function (52) aims to maximize the airline's profit, which is composed of revenue and flight operation costs. Constraints (53) and (54) are flight cover constraints for mandatory and optional flights, respectively. Constraints (55) ensure that every aircraft is assigned to at most one route. Routes starting from stations that are not the initial position of the aircraft constraints are prohibited as specified by Constraints (56). Constraints (57) ensure that the number of aircraft of the same fleet type starting at one airport is equal to that ending at that airport. Constraints (58) and (59) define aircraft routes from flight assignment decisions. Constraints (60) ensure that itineraries with a cancelled optional flight ought to be excluded. Otherwise, an itinerary with all its composing flights executed is included as regulated in Constraints (61). Constraints (62) relate the optional flights to the corresponding flights. Constraints (63) restrict the assigned passenger number on one flight to be no more than the number of available seats. Constraints (64) restrict the accepted passenger number to be no more than the demand of the itinerary. Constraints (65) ensure that the assigned passenger number on its own itinerary is zero if the itinerary is not activated. Lastly, the recaptured passenger number is limited by Constraints

Table 16. Additional notation for the integrated schedule design, fleet assignment and aircraft routing model M9.

Sets	
F^M	Set of mandatory flights.
F^O	Set of optional flights.
$F(i)$	Set of flights within itinerary i .
$F^O(i)$	Set of optional flights belonging to itinerary i .
AP	Set of airports, $ap \in AP$.
A^{ap}	Set of aircraft whose initial station is ap .
I^O	Set of itineraries involving any optional flight.
$I(f)$	Set of itineraries involving flight f .
I_i	Set of itineraries to which demand for itinerary i can be routed.
V	Set of vertices in vertex set of connection network; flight vertex is denoted by $v \leq n$ and airport vertex is denoted by $n + ap$.
E	Set of connecting edges in the connection network, $(v_1, v_2) \in E$.
Parameters	
T_{af}	Flying duration of flight f when it is assigned to aircraft a .
τ_a	Turn time of aircraft a .
$[lb_f, ub_f]$	Departure time window of flight f .
τ_a	Turn time of aircraft a .
Cap_a	Capacity of aircraft a .
$\beta_{i_1 i_2}$	Recapture rate of spilled demand for itinerary i_1 that is rerouted to itinerary i_2 .
Decision variables	
x_{af}	1 if aircraft a is assigned to flight f , 0 otherwise.
y_{a, v_1, v_2}	1 if aircraft a serves vertex v_2 immediately after vertex v_1 , 0 otherwise.
$\pi_{i_1 i_2}$	Number of passengers within itinerary i_1 who are accommodated within itinerary i_2 .
z_i	1 if itinerary i is selected for inclusion, 0 otherwise.

(66) if there are passengers not accommodated on their original itinerary.

6.3 Integrated aircraft routing and crew pairing problem

Research that integrates CSP with ARP provides feasible, efficient aircraft routing and crew pairing schedules jointly, compared with sequential models. In particular, such integration can reduce the operation cost induced by inappropriate crew routing (e.g. crew deadhead) and the crew switching aircraft. Instead of the traditional tactical purposes, [Ruther et al. \(2016\)](#) presented a model to re-optimize aircraft routes and crew pairings close to the day of operation and repair operation disruptions. In order to ensure tractability, an improved column generation method including superimposed pricing problems and early branching was developed to solve real-world instances. Targeting industrial problems in Air France, [Parmentier and Meunier \(2020\)](#) proposed a compact aircraft routing model and utilized a cut generation dynamic algorithm to solve the integrated aircraft routing and crew scheduling problem efficiently. In [Shao et al. \(2017\)](#), a compact formulation that incorporates aircraft fleetings, routing, and crew pairing operations was developed with the crew pairing part being modelled using the set partition formulation. This problem was solved using an accelerated Benders decomposition algorithm, and column generation with a perturbed Lagrangian dual was embedded as well. As part of the complete crew scheduling problem, the crew pairing and crew assignment subproblems were integrated into the work of [Zeighami et al. \(2020\)](#) to generate

better pairings for the vocation requests of crews. The computational difficulties were tackled using Benders decomposition and column generation and validated on the flight schedule of an American airline.

Finally, given the notation in [Table 17](#), we present the integrated aircraft routing and crew pairing model of [Ruther et al. \(2016\)](#) in Constraints (67)–(74):

$$(M10) \quad \min \sum_{b \in B} \sum_{p \in P_b} c^{pb} y^{pb} + \sum_{a \in A} \sum_{r \in R_a} c^{ra} x^{ra} + \sum_{(f_1, f_2) \in C_{Re}} \rho_{f_1 f_2} z_{f_1 f_2} \quad (67)$$

$$\sum_{a \in A} \sum_{r \in R_a} v_f^{ra} x^{ra} = 1 \quad \forall f \in F^1 \quad (68)$$

$$\sum_{b \in B} \sum_{p \in P_b} v_f^{pb} y^{pb} = 1 \quad \forall f \in F^1 \quad (69)$$

$$\sum_{b \in B} \sum_{p \in P_b} v_f^{pb} y^{pb} \leq 1 \quad \forall f \in F^0 \quad (70)$$

$$\sum_{r \in R_a} x^{ra} \leq 1 \quad \forall a \in A \quad (71)$$

$$\sum_{p \in P_b} y^{pb} \leq n_b \quad \forall b \in B \quad (72)$$

$$\sum_{b \in B} \sum_{p \in P_b} h_{f_1 f_2}^{pb} y^{pb} - \sum_{a \in A} \sum_{r \in R_a} h_{f_1 f_2}^{ra} x^{ra} \leq 0 \quad \forall (f_1, f_2) \in C_{Sh} \quad (73)$$

$$\sum_{b \in B} \sum_{p \in P_b} h_{f_1 f_2}^{pb} y^{pb} - \sum_{a \in A} \sum_{r \in R_a} h_{f_1 f_2}^{ra} x^{ra} - z_{f_1 f_2} \leq 0 \quad \forall (f_1, f_2) \in C_{Re}. \quad (74)$$

The objective function (67) minimizes the operation cost of the selected aircraft routes and crew pairings along with the penalty cost if crews change aircraft on a restricted connection. Constraints (68) and (69) require that one aircraft and crew must be assigned to every flight in period 1, which takes up the first four days in a seven-day problem. In contrast, Constraints (70) do not restrict every flight in period 2 (the remaining three days) to being covered by one crew, in order to ensure that crews can return to their respective bases. Constraints (71) and (72) limit the number of available aircraft and crews at each base station. Linking Constraints (73) requires that a short connection can be used by crews only if it is also covered by an aircraft. Constraints (74) further penalize aircraft changes on restricted connections.

6.4 Integrated fleet assignment and crew pairing problem

By jointly optimizing the fleet assignment and crew pairing problem, crew resources are more efficiently allocated to all the flights, in contrast to the sequential approach that derives the crew pairings based on predetermined fleets (families). In [Sandhu and Klabjan \(2007\)](#), a model integrating fleetings and crew pairing decision was proposed. Plane-count linking constraints were additionally incorporated to ensure feasibility. Both Benders decomposition and Lagrangian relaxation techniques were applied to achieve integrated solutions with a 3 per cent decrease in the overall planned cost. Targeting a European regional carrier, [Salazar-González \(2014\)](#) developed a compact arc-based integrated model that combines the fleet assignment, aircraft routing, and crew

Table 17. Additional notation for the integrated aircraft routing and crew pairing model M10.

Sets	
F^1	Set of flights in Period 1.
F^0	Set of flights in Period 2.
B	Set of crew blocks, $b \in B$.
P_b	Set of feasible pairings from crew block b .
C_{Sh}	Set of short connections where duration is smaller than minimum sit time, $(f_1, f_2) \in C_{Sh}$.
C_{Re}	Set of restricted connections where duration is less than 90 min, $(f_1, f_2) \in C_{Re}$.
Parameters	
v_f^a	1 if route r served by aircraft a covers flight f , 0 otherwise.
v_f^{pb}	1 if flight f is covered by pairing p crew block b , 0 otherwise.
c^{ra}	Cost of route r served by aircraft a .
c^{pb}	Cost of pairing p for crew block b .
n_b	Number of crews in crew block b .
$h_{f_1 f_2}^{ra}$	1 if connection $(f_1, f_2) \in C_{Sh} \cup C_{Re}$ is part of route r , aircraft a , 0 otherwise.
$h_{f_1 f_2}^{pb}$	1 if connection $(f_1, f_2) \in C_{Sh} \cup C_{Re}$ is part of pairing p , crew block b , 0 otherwise.
$\rho_{f_1 f_2}$	Penalty for restricted connection.
Decision variables	
x_{ra}	1 if route r is chosen for aircraft a , 0 otherwise.
y_{pb}	1 if pairing p is chosen for crew block b , 0 otherwise.
$z_{f_1 f_2}$	1 if a restricted connection is used by a crew but not by an aircraft, 0 otherwise.

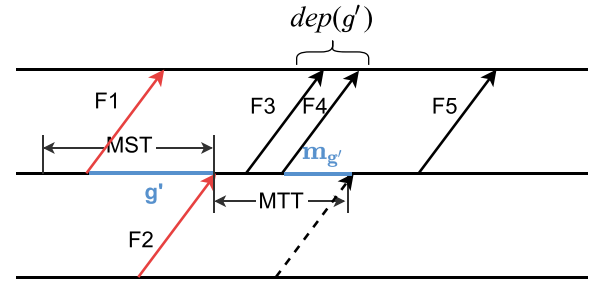
pairing problems without daylight aircraft maintenance. To integrate fleet assignment, aircraft routing, and crew scheduling, [Cacchiani and Salazar-González \(2017\)](#) presented two MIP models for a regional airline. Aiming at deriving exact solutions of the two models, a three-phase solution algorithm combining bounding cut and branch and price was proposed. In [Ahmed et al. \(2022\)](#), an integrated robust optimization problem containing the last three subproblems was solved by presenting a MINP arc-based model. Robustness was constructed by penalizing short connections and encouraging crews to follow aircraft routes. To handle the computational intractability, the model was decomposed and solved according to different aircraft families and improved using a proximity search method.

The model formulation of [Sandhu and Klabjan \(2007\)](#) introduced in this part is based on [Table 18](#) to highlight the key modifications for integration between fleet assignment and crew pairing decisions. In particular, this model relies on *essential ground arcs* to capture the consistency between aircraft and crew routes when the connection time for a flight pair is less than the minimum sit time (MST), which is typically larger than the minimum turnaround time (MTT). [Fig. 6](#) illustrates an essential ground arc g' which is defined as a departure (F1) followed by an arrival (F2); their interval is shorter than the MST. While flight F2 can only connect successive flights after an additional MTT (i.e. after F4), the number of crews passing arc g' should not exceed the accumulated aircraft number, which is the summation of departure flight number $|\text{dep}(g')|$ and on-ground aircraft number $z_{m_g'k}$:

$$(M11) \quad \max \sum_{k \in K} \sum_{f \in F} r_{kf} x_{kf} - \sum_{k \in K} \sum_{p \in P} c_p y_{kp} \quad (75)$$

Table 18. Additional notation for the integrated fleet assignment and crew pairing model M11.

Sets	
V	Set of vertices in the time-space network, $v \in V$.
$E(g')$	Set of essential ground arcs.
$FI(v)$	Set of flights to node v .
$FO(v)$	Set of flights from node v .
FC	Set of flights crossing the count time.
W	Set of ground arcs crossing the count time.
Q	Set of all fleet families, $q \in Q$.
P_g	Set of pairings that have a forced turn (connection time less than MST) through g .
P_f	Set of pairings covering flight f .
$\text{dep}(g')$	Set of departing flights associated with g' .
$i(v)/o(v)$	Ground arc to/from node v .
Parameters	
r_{kf}	Profit of assigning fleet k to flight f .
c_p	Cost of a given pairing p .
a_{pg}	Number of times pairing p includes arc g .
$h(k)$	Corresponding fleet family of fleet k .
Decision variables	
x_{kf}	1 if flight f is assigned to fleet type k , 0 otherwise.
y_{kp}	1 if pairing p is assigned to fleet k , 0 otherwise.
y_{qp}	1 if pairing p is assigned to fleet family q , 0 otherwise.
z_{gk}	Number of aircraft of fleet type k on ground arc g .

**Figure 6** Example of an essential ground arc.

$$\sum_{k \in K} x_{kf} = 1 \quad \forall f \in F \quad (76)$$

$$\sum_{f \in FO(v)} x_{kf} + z_{o(v)k} - \sum_{f \in FI(v)} x_{kf} - z_{i(v)k} = 0 \quad \forall v \in V, \forall k \in K \quad (77)$$

$$\sum_{f \in FC} x_{kf} + \sum_{g \in W} z_{gk} \leq N_k \quad \forall k \in K \quad (78)$$

$$\sum_{p \in P_f} y_{h(k)p} = x_{kf} \quad \forall f \in F, \forall k \in K \quad (79)$$

$$\sum_{p \in P(g')} a_{pg'} y_{qp} \leq \sum_{k \in S_q} (z_{m_g'k} + \sum_{f \in \text{dep}(g')} x_{kf}) \quad \forall g' \in E(g'), q \in Q. \quad (80)$$

The objective function (75) tries to maximize the fleet profit and reduce the crew pairing cost. Flight cover constraints (76) and flow balance constraints (77) ensure that every flight is served

by one fleet. Constraints (78) limit the available aircraft number by selecting a count timestamp. Constraints (79) ensure that a pairing is assigned to a fleet family if all the component flights are assigned to the same family. Lastly, the plane-count constraints (80) are introduced for each fleet family and essential ground arc.

To conclude this section, the integration of multiple subproblems contributes to improved flight/crew schedules featuring higher profits, lower operation cost, and reduced conflicts between different planning phases. However, so far, there has been little work aiming at solving the scheduling problem that integrates all four subproblems, possibly due to two factors. First, the intrinsic high complexity induced by a fully integrated model poses great challenges for solving real-world instances. Second, the possible conflicts caused by different planning horizons of the four subproblems may incur a transformation in airline operation management procedures. These also provide new insights for further studies of more complicated models and efficient algorithms. Besides, robustness concerning both aircraft routing and crew scheduling is worth refined consideration in terms of joint delay propagations.

7. Robust scheduling problems

While the aforementioned airline planning process is usually carried out under the assumption that operations will be executed as planned, the airline industry is susceptible to external changes or disruptions, such as demand variations, convective weather, and congested airspace. As a result, the study of the robust airline scheduling problem has garnered increasing interest in recent research studies, aiming to mitigate potential financial losses caused by disruptions. Whilst robustness-related works have been referred to in some previous sections, this section focuses on presenting the corresponding specific model formulations for demand- and delay-related uncertainties. The relevant literature is summarized in Table 19.

7.1 Demand fluctuation

As one of the most important factors that determine the airline flight schedule, demand is usually considered months before the day of operation in conjunction with FAP in order to facilitate the decisions on aircraft and crew scheduling. However, market demand is still highly uncertain at that time and can pose a significant impact on the overall profit level of flight schedules. To handle such challenges, rescheduling decisions may be made with a relatively tight planning horizon (Jiang and Barnhart 2013) based on updated demand predictions with high reliability. Besides, robustness/uncertainty-related modelling methodologies have also been closely studied (Sherali and Zhu, 2008; Cadorso and Celis, 2017; Kenan et al. 2018). In this part, the stochastic programming model of Cadorso and Celis (2017) is presented, which incorporates a set of passenger demand scenarios with corresponding weights. The notation is explained in Table 20:

$$(M12) \max \sum_{s \in S} \sum_{i \in I} \omega_s (\text{fare}_i - \text{mpc}_i \text{prob}_i) h_i^s - \sum_{f \in F} \sum_{k \in K} c_{fk} z_f^k \quad (81)$$

$$\sum_{i \in I} h_i^s \leq d_w^s - \sum_{i_2 \in I} t_{i_2}^i + \sum_{i_2 \in I} b_{i_2}^i t_{i_2}^i \quad \forall i \in I, s \in S \quad (82)$$

$$\sum_{i \in I} (1 - \delta_f^i \text{prob}_i) h_i^s \leq \sum_{k \in K} \text{Cap}_k z_f^k \quad \forall f \in F, s \in S \quad (83)$$

$$\sum_{k \in K} z_f^k = 1 \quad \forall f \in F_M \quad (84)$$

$$\sum_{k \in K} z_f^k \leq 1 \quad \forall f \in F_O \quad (85)$$

$$y_v^p + \sum_{f \in AS_{v,k}} z_f^p = y_v^p + \sum_{f \in DS_{v,k}} z_f^k \quad \forall v \in V, k \in K \quad (86)$$

$$\sum_{f \in F} \beta_f z_f^k + \sum_{v \in V} \gamma_v y_v^k \leq N_k \quad \forall k \in K \quad (87)$$

$$y_{v_i}^k = y_{v_j}^k \quad \forall ap \in AP, k \in K \quad (88)$$

$$\frac{h_i^s}{d_w^s} - \sum_{i_2 \in I} t_{i_2}^i \leq \sum_{k \in K} z_f^k \quad \forall f \in F, i \in I_f, s \in S. \quad (89)$$

The objective function (81) aims to maximize the airline's profit, considering revenue and missed connections in the first term and flight operation cost in the second term. Due to different demand realizations, a weighted cost is derived for all scenarios. Constraints (82) compute for each scenario the number of passengers carried in every itinerary including spilled and recaptured passenger flow. Constraints (83) ensure that the number of passengers on a flight does not exceed the aircraft capacity. Missed connections are taken into account as well for one-stop itineraries. Constraints (84) and (85) are flight cover constraints for mandatory flights and optional flights, respectively. Network flow balance is enforced by Constraints (86) where the grounded aircraft number at a vertex is the grounded aircraft before the vertex minus departing aircraft plus arriving aircraft. Aircraft count constraints (87) ensure that the number of aircraft used at any time does not exceed the available number. Constraints (88) serve as wrap-around edges linking the last and first events for every airport. Lastly, Constraints (89) are valid inequalities that tighten the linear relaxation of the model. Specifically, when h_i^s is relatively small compared to the available seats of the assigned aircraft, if the quotient is 1 and there is no spilled passenger, it will force $\sum_{k \in K} z_f^k = 1$.

The stochastic programming model M9 employs an L-shaped structure and is solved using Benders decomposition. The Benders master problem consists of variables z_f^k, y_v^k and Constraints (55)–(59). The subproblem for each scenario is composed of Constraints (53), (54) and (60). While the two-phase Benders decomposition method is applied in this model, the generated Benders cuts in the first phase, where the master problem is relaxed into a linear programming model, are aggregated into a single constraint before the start of the second phase.

7.2 Propagated delays

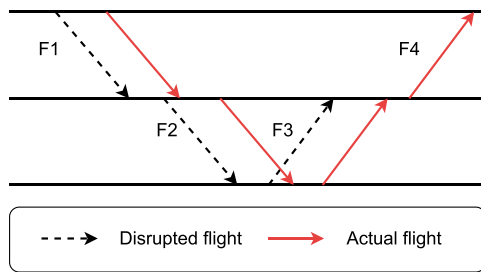
Another frequently studied robustness feature in the airline scheduling field is propagated delays, which are caused by inappropriate allocation of aircraft/crew turnaround time as illustrated in Fig. 7. In order to mitigate the significant costs induced by delays concerning crew salary, passenger dissatisfaction, and aircraft utilization, robust aircraft/crew routing methods are proposed and developed (Lan et al. 2006; Aloulou et al. 2013; Liang et al. 2015).

As previously introduced in ARP, related literature can be classified as recoverability and stability-based approaches. For the recoverability approach, Rosenberger et al. (2004) derived fleet assignment solutions with shorter aircraft cycles, which will provide more flexibility towards cancellation decisions. By integrating the aircraft recovery subproblem, Froyland et al. (2014) presented a tail assignment model (operational aircraft routing)

Table 19. Summary of the literature for robust airline scheduling.

Literature	SDP	FAP	ARP	CSP	Method	Flight	Solution time	Robust feature
Rosenberger et al. (2004)		✓			B&B	1,200	–	Short cycles
Lan et al. (2006)			✓		B&P	278	13 s	Delay propagation, passenger connections
Yen and Birge (2006)				✓	B&P	79	–	Crew switching planes
Sherali and Zhu (2008)		✓			BD	900	6.8 h	Demand uncertainty
Weide et al. (2010)			✓	✓	CG, iterative algorithm	750	2,926 s	Delay propagation
Sohoni et al. (2011)	✓				Row generation	1,500	3,921 s	Block-time uncertainty
Dück et al. (2012)			✓	✓	CG, dynamic programming	272	3.3 h	Propagated delays
Dunbar et al. (2012)			✓	✓	CG, iterative algorithm	54	47.19 s	Delay propagation
Aloulou et al. (2013)	✓		✓		B&B	1,278	786.91 s	Delay propagation
Froyland et al. (2014)			✓		B&P	53	3,093.93 s	Delay propagation
Jiang and Barnhart (2013)	✓	✓			CG	1,000	10 h	Demand variability
Maher et al. (2014)			✓		BD	3,370	65,087 s	Maintenance misalignment
Liang et al. (2015)			✓		CG	6,072	10,059 s	Delay propagation
Gürkan et al. (2016)	✓	✓	✓		Multistage search	114	3254.3 s	Delay propagation
Ahmed et al. (2017)	✓		✓		B&B	1,195	3,600 s	Delay and passengers
Cadarso and Celis (2017)	✓	✓			BD	–	3841 s	Demand uncertainty
Kenan et al. (2018)	✓	✓	✓		CG	228	6854 s	Demand uncertainty, delay propagation
Maher et al. (2018)			✓		B&P, iterative algorithm	3,370	76.8 s	Maintenance misalignment
Marla et al. (2018)			✓		CG, depth-first search	165	6,018.39 s	Delay propagation
Şafak et al. (2018)	✓	✓	✓		SOCP, cutting plane	102	17.7 h	Demand and non-cruise time uncertainty
Antunes et al. (2019)				✓	CG	94	14,400 s	Delay propagation
Cacchiani and Salazar-González (2020)		✓	✓	✓	CG, preselection heuristic	150	4,397 s	Connection time
Xu et al. (2021)	✓	✓	✓		CG, variable neighbourhood search	150	4,397 s	Connection time

CG: column generation; BD: Benders decomposition; B&B: branch and bound; B&P: branch and price; –: not reported.

**Figure 7** Flight delay propagation on an aircraft (crew) route.

that minimizes the weighted recovery cost using a stochastic programming approach. Concerning the stability approach, Liang et al. (2015) presented a robust model based on a weekly line-of-flight network and optimized the expected propagated delays of aircraft routes to derive robust weekly aircraft routing solutions. Due to the non-linearity computation of propagated delays, column generation was adopted as the solution methodology. In Antunes et al. (2019), a robust optimization model based on a budget uncertainty set was developed to solve the crew pairing problem and reduce delay propagation via crew connections.

In this section, the mathematical formulation of Antunes et al. (2019) is presented; this aims to reduce the expected propagated delays of crew pairings. With the relevant notation defined in Table 21, the objective function and constraints are displayed as follows:

$$(M13) \quad \min \sum_{p \in P} \eta_p + \gamma (\Gamma \zeta + \sum_{f \in F} q_f) \quad (90)$$

$$\sum_{p \in P} k_{fp} x_p = 1 \quad \forall f \in F \quad (91)$$

$$\zeta = \sum_{p \in P} \zeta_p \quad (92)$$

$$q_f = \sum_{p \in P} q_{fp} \quad \forall f \in F \quad (93)$$

$$\zeta_p + q_{fp} \geq \sum_{f' \in F} p_{f'fp} + \Delta_f k_{fp} x_p \quad \forall f \in F, \forall p \in P \quad (94)$$

$$p_{f_1 f_2 p} \geq \sum_{f' \in F} p_{f' f_1 p} + \Delta_{f_1} x_p - (S_{f_1 f_2} - \tau) \quad \forall (f_1, f_2) \in L_p^s, \forall p \in P \quad (95)$$

Table 20. Additional notation for the robust airline scheduling model M12 with demand uncertainty.

Sets	
S	Set of scenarios, $s \in S$.
AP	Set of airports, $ap \in AP$.
V	Set of vertices, $v \in V$. Each vertex is defined by an airport and a time period. v^- denotes vertex before v and v_i^{ap}, v_f^{ap} denote the initial and final vertices for airport ap , respectively.
I_f	Subset of itineraries using flight f .
$DS_{v,k}$	Subset of flights that depart from vertex v with fleet type k assigned.
$AS_{v,k}$	Subset of flights that arrive in vertex v with fleet type k assigned.
F_M	Set of mandatory flights that must be executed.
F_O	Set of optional flights that can be cancelled.
Parameters	
ω_s	Weight of scenario s .
$b_{i_1}^{i_2}$	recapture rate from itinerary i_1 to i_2 .
$prob_i$	Probability of getting misconnected passengers in itinerary i .
β_f	1 if flight f is flying during the count time, 0 otherwise.
γ_v	1 if vertex v contains the count time.
D_i^s	Passenger group size in itinerary i and scenario s .
δ_f^j	1 if i is a one-stop itinerary and f is the second flight in the itinerary.
Decision variables	
z_f^k	1 if flight f is assigned fleet type k , 0 otherwise.
y_v^k	Number of aircraft of fleet type k grounded in vertex v .
h_i^s	Number of passengers travelling in itinerary i in scenario s .
$t_{i_1}^{i_2}$	Number of passengers requesting itinerary i_1 but who are redirected to itinerary i_2 .

$$p_{f_1 f_2 p} \geq \sum_{f' \in F} p_{f' f_1 p} + \Delta_{f_1} x_p - (\tau_{f_1 f_2} - \rho)$$

$$\tau_{f_1 f_2} - \rho) \quad \forall (f_1, f_2) \in L_p^r, \forall p \in P \quad (96)$$

$$\eta_p \geq \sum_{d \in D_p} \psi_{dp} \quad \forall p \in P \quad (97)$$

$$\eta_p \geq \alpha(T_p x_p + (\eta_p + q_{f_p}^s)) \quad \forall p \in P \quad (98)$$

$$\psi_{dp} \geq b_{dp} x_p \quad \forall d \in D, \forall p \in P \quad (99)$$

$$\psi_{dp} \geq \beta(e_{dp} x_p + (\zeta_p + q_{f_{dp}}^s)) \quad \forall d \in D, \forall p \in P \quad (100)$$

$$\psi_{dp} \geq \theta x_p \quad \forall d \in D, \forall p \in P \quad (101)$$

$$\lambda \geq e_{dp} x_p + (\zeta_p + q_{f_{dp}}^s) \quad \forall d \in D. \quad (102)$$

The objective function (90) seeks to minimize the crew pairing and flight delay cost due to propagated delays. The robustness budget parameter is Γ , where at most $\lfloor \Gamma \rfloor$ flights take their worst-case value Δ_f and one other flight takes a value that is $\Gamma - \lfloor \Gamma \rfloor$ times as bad as the worst-case value. Constraints (91) ensure that every flight is covered by one pairing. Constraints (92) and (93) compute the delay values of ζ and q_f from their pairing counterparts. Constraints (94) capture the total arrival delay of flight f in pairing p considering its root delay and propagated delay. Apparently, when Γ takes a large value, Γ_p tends to decrease and the more extreme delay Δ_f will be incorporated by variable

Table 21. Additional notation for the robust crew pairing model M13.

Sets	
L_p^s	Set of ordered flight pairs such that the second flight in the pair immediately follows the first within the same duty in pairing p .
L_p^r	Set of ordered flight pairs such that the second flight in the pair immediately follows the first, but with the two being in consecutive duties in pairing p .
D_p	Set of all duties in pairing p , $d \in D_p$.
Parameters	
Γ	Robustness budget $\Gamma \in [0, F]$.
k_{fp}	1 if flight f is included in pairing p .
γ	Ratio of the cost of one hour of flight delay to the crew pay for one hour of flying time.
v_p	Number of flights in pairing p .
α, β, θ	Parameters in pay-and-credit crew salary scheme.
b_{dp}	Sum of planned block time of flights in duty d , pairing p .
Δ_f	Root delay of flight f .
e_{dp}	Planned elapsed time of duty d in pairing p .
T_p	Time away from base of pairing p .
$s_{f_1 f_2}$	Planned sit time between flights f_1 and f_2 .
$r_{f_1 f_2}$	Planned rest time between flights f_1 and f_2 .
f_d	Last flight in duty d .
$\bar{f}_d$	Last flight in pairing p .
τ	Minimum sit time.
ρ	Minimum rest time.
λ	Maximum duty elapsed time.
Decision variables	
x_p	1 if pairing p is chosen, 0 otherwise.
ζ	Auxiliary delay variable.
q_f	Auxiliary delay variable associated with flight f .
ζ_p	Auxiliary delay variable associated with pairing p .
q_{fp}	Auxiliary delay variable associated with flight f and pairing p .
η_p	Robust version of the cost of pairing p .
ψ_{dp}	Robust version of the cost of duty d in pairing p .
$p_{f_1 f_2 p}$	Propagated delay from flight f_1 to flight f_2 within pairing p .

q_{fp} . Constraints (95) and (96) compute the propagated delay in a pairing considering different buffer times of flight pairs (duty flight connections and overnight rests). Constraints (97)–(101) jointly model the pay-and-credit crew salary scheme. Specifically, Constraints (97) ensure that the total cost is no less than the summation of duty costs. Constraints (98) compute the cost associated with time away from base. Constraints (99)–(101) calculate the lower bounds corresponding to duty block time, elapsed time, and the minimum guaranteed crew pay. Finally, Constraints (102) limit the maximum elapsed time of every duty.

In sum, with the increase in air traffic flow, traditional models that solely optimize cost or revenue may result in fragile flight schedules regarding the fluctuation of critical input parameters such as demand and flight departure/arrival time. Therefore, there is growing interest in robust airline scheduling problems using either domain knowledge (e.g. short aircraft cycles in [Rosenberger et al. \(2004\)](#)) or stochastic/robust optimization techniques (e.g. [Cadarsó and Celis \(2017\)](#); [Antunes et al. \(2019\)](#)). Accordingly, the high complexity of such robust models usually requires customized solution algorithms (e.g. column generation and branch and cut in [Yan and Kung \(2018\)](#)) to handle uncertainty sets and a set of scenarios.

8. Conclusion

In this paper, we have reviewed the airline scheduling optimization problem and its various modelling methodologies. Traditionally, the problem is divided into four subproblems: schedule design, fleet assignment, aircraft routing, and crew scheduling (including crew pairing and crew rostering) to alleviate computational complexity and facilitate practical implementation across different airline departments. Several types of mathematical models, such as the multi-commodity flow model and the set partition model, are also developed based on different spatio-temporal network structures that correspond to varying sizes, densities, and abilities of handling realistic operational constraints.

Nevertheless, intensified market competition drives the study for integrated models that combine two or three subproblems. By incorporating the interrelationship between aircraft, crews, and passengers, integrated models have shown promising results in real-world case studies to achieve flight schedules with a significant decrease in operational cost and inconsistency. The relevant model formulation either chooses to aggregate overnight flight connections or adopt a set partition formulation to match the surge in model size. Although extensive efforts have been devoted in the literature to the development of scalable algorithms, solving medium- to large-scale integrated models remains a challenge, especially when using near-exact decomposition algorithms such as column generation. Heuristic algorithms, on the other hand, are less preferred, possibly due to the lack of optimality gap guarantees and the complicated problem-dependent tuning techniques.

Moreover, in light of escalating traffic flow and frequently encountered operational irregularities, multiple studies delve into robust scheduling problems in order to enhance the robustness or resilience of air transportation, including in multimodal contexts (Wandelt et al. 2023). To efficiently incorporate such features, researchers have designed domain-knowledge-based measurements (e.g. station purity), but these may not be easily generalized to other scheduling problems with satisfying performances. Robust optimization and stochastic programming techniques have also been proposed with various scenarios and uncertainty sets, respectively. Capturing the empirical information using Benders decomposition, column generation, and cutting plane methods provides effective guidance for fleet capacity allocation and critical flight protection whilst unavoidably bringing additional computational difficulties.

Based on the findings from our review of the models and literature, several directions of future research can be discussed. First, regarding the integration of different subproblems, a more sophisticated coupling relationship should be modelled to enhance profitability and punctuality performance. For instance, although demand–supply interaction is frequently studied in the extant literature for the integrated schedule design and fleet assignment problem (Atasoy et al. 2014; Wei et al. 2020), the potential impact of frequency, passenger connection time, and airline competition needs to be explicitly modelled, incorporating more accurate network effects. Second, to handle the side effects of parameter uncertainty, both adaptive scheduling models based on updated demand information (Yan and Ghatge 2022) and robust models that incorporate historical statistics ought to be constructed. On the other hand, although delay propagation has attracted more studies to optimize aircraft routing/crew pairings with different robustness measurements (Marla et al. 2018), the correlations between these two entities are expected to be further optimized to

appropriately reduce the inconsistency between aircraft and crew routes for flights with high delay risks. Specifically, when computing the expected propagated delay induced by aircraft routes, the secondary delay incurred by inappropriate crew turnarounds is typically not explicitly computed in the current literature to reach a trade-off between operating efficiency and on-time performance. Additionally, schedule planning can be further extended towards aircraft turnaround operation (e.g. stand/gate assignment) and the multiple airport system (e.g. correlated runway/slot capacity) to enable more accurate delay/cost estimation and efficient operations.

Regarding computational tractability, large-scale integrated airline scheduling problems remain challenging to solve using traditional exact or near-exact solution algorithms due to rapidly increasing solution spaces and extensive interactions between operating resources. For instance, it takes up to 260 s to solve the fleet assignment subproblem model M1 for an instance containing 1,400 flights and 1,786 passenger itineraries with an i7-6500U CPU. In contrast, the solution time for the integrated model M8, which combines schedule design, fleet assignment, and aircraft routing decisions, rises rapidly to 180 min under the same computational setting. Moreover, stochastic programming or robust optimization models may require the incorporation of subproblems with high complexity to dynamically add cuts or variables. While the deterministic model M8 is solved within 340 min, including demand scenarios in model M12 leads to an additional solution time of 360 min. As such, further research should focus on utilizing efficient searching heuristics (Xu et al. 2021) and machine learning techniques (Tahir et al. 2021) to successfully prune solution spaces and finding high-quality initial solutions. It is also worth mentioning that the difference in planning horizons, operational requirements, and modelling granularity (e.g. leg-based demand vs itinerary-based demand) poses difficulties in fairly comparing the efficiency and performance of varying methodologies in the literature. Therefore, benchmark datasets and algorithmic implementations covering the aforementioned scheduling problems with necessary adaptations (e.g. a daily problem extends to a weekly problem with schedule synchronization constraints) are expected to be developed to facilitate successive validation and reproducibility.

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References

- Abara, J. (1989) 'Applying Integer Linear Programming to the Fleet Assignment Problem', *Interfaces*, **19**: 20–8.
- Ahmed, M. B. et al. (2017) 'A Hybrid Optimization-Simulation Approach for Robust Weekly Aircraft Routing and Retiming', *Transportation Research Part C: Emerging Technologies*, **84**: 1–20.
- Ahmed, M. B. et al. (2022) 'A Matheuristic for the Robust Integrated Airline Fleet Assignment, Aircraft Routing, and Crew Pairing Problem', *Computers & Operations Research*, **137**: 105551.
- Ahuja, R. K. et al. (2007) 'A Very Large-Scale Neighborhood Search Algorithm for the Combined through-Fleet-Assignment Model', *INFORMS Journal on Computing*, **19**: 416–28.
- Aloulou, M. A., Haouari, M., and Mansour, F. Z. (2013) 'A Model for Enhancing Robustness of Aircraft and Passenger Connections', *Transportation Research Part C: Emerging Technologies*, **32**: 48–60.
- Antunes, D., Vaze, V., and Antunes, A. P. (2019) 'A Robust Pairing Model for Airline Crew Scheduling', *Transportation Science*, **53**: 1751–71.
- Atasoy, B., Salani, M., and Bierlaire, M. (2014) 'An Integrated Airline Scheduling, Fleet, and Pricing Model for a Monopolized Market', *Computer-Aided Civil and Infrastructure Engineering*, **29**: 76–90.
- Barnhart, C., Farahat, A., and Lohatepanont, M. (2009) 'Airline Fleet Assignment with Enhanced Revenue Modeling', *Operations Research*, **57**: 231–44.
- Barnhart, C., Hatay, L., and Johnson, E. L. (1995) 'Deadhead Selection for the Long-Haul Crew Pairing Problem', *Operations Research*, **43**: 491–9.
- Barnhart, C., Kniker, T. S., and Lohatepanont, M. (2002) 'Itinerary-Based Airline Fleet Assignment', *Transportation Science*, **36**: 199–217.
- Barnhart, C. et al. (1998) 'Flight String Models for Aircraft Fleet and Routing', *Transportation Science*, **32**: 208–20.
- Başdere, M., and Bilge, Ü. (2014) 'Operational Aircraft Maintenance Routing Problem with Remaining Time Consideration', *European Journal of Operational Research*, **235**: 315–28.
- Bélanger, N. et al. (2006a) 'Periodic Airline Fleet Assignment with Time Windows, Spacing Constraints, and Time Dependent Revenues', *European Journal of Operational Research*, **175**: 1754–66.
- Bélanger, N. et al. (2006b) 'Weekly Airline Fleet Assignment with Homogeneity', *Transportation Research Part B: Methodological*, **40**: 306–18.
- Berge, M. E., and Hopperstad, C. A. (1993) 'Demand Driven Dispatch: a Method for Dynamic Aircraft Capacity Assignment, Models and Algorithms', *Operations Research*, **41**: 153–68.
- Birolini, S. et al. (2021) 'Integrated Flight Scheduling and Fleet Assignment with Improved Supply-Demand Interactions', *Transportation Research Part B: Methodological*, **149**: 162–80.
- Burke, E. K. et al. (2010) 'A Multi-Objective Approach for Robust Airline Scheduling', *Computers & Operations Research*, **37**: 822–32.
- Cacchiani, V., and Salazar-González, J.-J. (2017) 'Optimal Solutions to a Real-World Integrated Airline Scheduling Problem', *Transportation Science*, **51**: 250–68.
- Cacchiani, V., and Salazar-González, J.-J. (2020) 'Heuristic Approaches for Flight Retiming in an Integrated Airline Scheduling Problem of a Regional Carrier', *Omega*, **91**: 102028.
- Cadarso, L., and de Celis, R. (2017) 'Integrated Airline Planning: Robust Update of Scheduling and Fleet Balancing under Demand Uncertainty', *Transportation Research Part C: Emerging Technologies*, **81**: 227–45.
- Clarke, L. et al. (1997) 'The Aircraft Rotation Problem', *Annals of Operations Research*, **69**: 33–46.
- Cui, R., Dong, X., and Lin, Y. (2019) 'Models for Aircraft Maintenance Routing Problem with Consideration of Remaining Time and Robustness', *Computers & Industrial Engineering*, **137**: 106045.
- Dawid, H., König, J., and Strauss, C. (2001) 'An Enhanced Rostering Model for Airline Crews', *Computers & Operations Research*, **28**: 671–88.
- Deveci, M., and Demirel, N. C. (2018) 'A Survey of the Literature on Airline Crew Scheduling', *Engineering Applications of Artificial Intelligence*, **74**: 54–69.
- Ding, Y. et al. (2023) 'Towards Efficient Airline Disruption Recovery with Reinforcement Learning', *Transportation Research Part E: Logistics and Transportation Review*, **179**: 103295.
- Doi, T., Nishi, T., and Voß, S. (2018) 'Two-Level Decomposition-Based Matheuristic for Airline Crew Rostering Problems with Fair Working Time', *European Journal of Operational Research*, **267**: 428–38.
- Dück, V. et al. (2012) 'Increasing Stability of Crew and Aircraft Schedules', *Transportation Research Part C: Emerging Technologies*, **20**: 47–61.
- Dunbar, M., Froyland, G., and Wu, C.-L. (2012) 'Robust Airline Schedule Planning: Minimizing Propagated Delay in an Integrated Routing and Crewing Framework', *Transportation Science*, **46**: 204–16.
- Eltoukhy, A. E., Chan, F. T., and Chung, S. H. (2017) 'Airline Schedule Planning: A Review and Future Directions', *Industrial Management & Data Systems*, **117**: 1201–43.
- Faust, O., Gonsch, J., and Klein, R. (2017) 'Demand-Oriented Integrated Scheduling for Point-to-Point Airlines', *Transportation Science*, **51**: 196–213.
- Froyland, G., Maher, S. J., and Wu, C.-L. (2014) 'The Recoverable Robust Tail Assignment Problem', *Transportation Science*, **48**: 351–72.
- Gopalan, R., and Talluri, K. T. (1998) 'The Aircraft Maintenance Routing Problem', *Operations Research*, **46**: 260–71.
- Guo, Y. et al. (2006) 'A Partially Integrated Airline Crew Scheduling Approach with Time-Dependent Crew Capacities and Multiple Home Bases', *European Journal of Operational Research*, **171**: 1169–81.
- Gürkan, H., Gürel, S., and Aktürk, M. S. (2016) 'An Integrated Approach for Airline Scheduling, Aircraft Fleet and Routing with Cruise Speed Control', *Transportation Research Part C: Emerging Technologies*, **68**: 38–57.
- Hane, C. A. et al. (1995) 'The Fleet Assignment Problem: Solving a Large-Scale Integer Program', *Mathematical Programming*, **70**: 211–32.
- Haouari, M., Shao, S., and Sherali, H. D. (2013) 'A Lifted Compact Formulation for the Daily Aircraft Maintenance Routing Problem', *Transportation Science*, **47**: 508–25.
- Haouari, M., Zeghal Mansour, F., and Sherali, H. D. (2019) 'A New Compact Formulation for the Daily Crew Pairing Problem', *Transportation Science*, **53**: 811–28.
- Haouari, M. et al. (2011) 'Exact Approaches for Integrated Aircraft Fleet and Routing at Tunisair', *Computational Optimization and Applications*, **49**: 213–39.
- Hoffman, K. L., and Padberg, M. (1993) 'Solving Airline Crew Scheduling Problems by Branch-and-Cut', *Management Science*, **39**: 657–82.
- International Air Transport Association (2023a) *Airline Profitability Outlook Strengthens*, accessed 25 June 2023. <https://www.iata.org/en/pressroom/2023-releases/2023-06-05-01/>.
- International Air Transport Association (2023b) *Industry Statistics Fact Sheet*, accessed 25 June 2023. <https://www.iata.org/en/iata-repository/pressroom/fact-sheets/industry-statistics/>.

- International Civil Aviation Organization (2020) *The World of Air Transport in 2019*, accessed 25 June 2023. <https://www.icao.int/annual-report-2019/Pages/the-world-of-air-transport-in-2019.aspx>.
- Ioachim, I. et al. (1999) 'Fleet Assignment and Routing with Schedule Synchronization Constraints', *European Journal of Operational Research*, **119**: 75–90.
- Jacobs, T. L., Smith, B. C., and Johnson, E. L. (2008) 'Incorporating Network Flow Effects into the Airline Fleet Assignment Process', *Transportation Science*, **42**: 514–29.
- Jiang, H., and Barnhart, C. (2009) 'Dynamic Airline Scheduling', *Transportation Science*, **43**: 336–54.
- Jiang, H., and Barnhart, C. (2013) 'Robust Airline Schedule Design in a Dynamic Scheduling Environment', *Computers & Operations Research*, **40**: 831–40.
- Kasirzadeh, A., Saddoune, M., and Soumis, F. (2017) 'Airline Crew Scheduling: Models, Algorithms, and Data Sets', *EURO Journal on Transportation and Logistics*, **6**: 111–37.
- Kenan, N., Jebali, A., and Diabat, A. (2018) 'The Integrated Aircraft Routing Problem with Optional Flights and Delay Considerations', *Transportation Research Part E: Logistics and Transportation Review*, **118**: 355–75.
- Khaled, O. et al. (2018a) 'A Compact Optimization Model for the Tail Assignment Problem', *European Journal of Operational Research*, **264**: 548–57.
- Khaled, O. et al. (2018b) 'A Multi-Criteria Repair/Recovery Framework for the Tail Assignment Problem in Airlines', *Journal of Air Transport Management*, **68**: 137–51.
- Lan, S., Clarke, J.-P., and Barnhart, C. (2006) 'Planning for Robust Airline Operations: Optimizing Aircraft Routings and Flight Departure Times to Minimize Passenger Disruptions', *Transportation Science*, **40**: 15–28.
- Lavoie, S., Minoux, M., and Odier, E. (1988) 'A New Approach for Crew Pairing Problems by Column Generation with an Application to Air Transportation', *European Journal of Operational Research*, **35**: 45–58.
- Liang, Z., and Chaovalitwongse, W. A. (2013) 'A Network-Based Model for the Integrated Weekly Aircraft Maintenance Routing and Fleet Assignment Problem', *Transportation Science*, **47**: 493–507.
- Liang, Z. et al. (2011) 'On a New Rotation Tour Network Model for Aircraft Maintenance Routing Problem', *Transportation Science*, **45**: 109–20.
- Liang, Z. et al. (2015) 'Robust Weekly Aircraft Maintenance Routing Problem and the Extension to the Tail Assignment Problem', *Transportation Research Part B: Methodological*, **78**: 238–59.
- Liu, M., Liang, B., Zheng, F., Chu, F., 11 (2018) 'Stochastic Airline Fleet Assignment with Risk Aversion', *IEEE Transactions on Intelligent Transportation Systems*, **20**: 1–10.
- Lohatepanont, M., and Barnhart, C. (2004) 'Airline Schedule Planning: Integrated Models and Algorithms for Schedule Design and Fleet Assignment', *Transportation Science*, **38**: 19–32.
- Maenhout, B., and Vanhoucke, M. (2010) 'A Hybrid Scatter Search Heuristic for Personalized Crew Rostering in the Airline Industry', *European Journal of Operational Research*, **206**: 155–67.
- Maher, S. J., Desaulniers, G., and Soumis, F. (2014) 'Recoverable Robust Single Day Aircraft Maintenance Routing Problem', *Computers & Operations Research*, **51**: 130–45.
- Maher, S. J., Desaulniers, G., and Soumis, F. (2018) 'The Daily Tail Assignment Problem under Operational Uncertainty Using Look-Ahead Maintenance Constraints', *European Journal of Operational Research*, **264**: 534–47.
- Makri, A., and Klabjan, D. (2004) 'A New Pricing Scheme for Airline Crew Scheduling', *INFORMS Journal on Computing*, **16**: 56–67.
- Marla, L., Vaze, V., and Barnhart, C. (2018) 'Robust Optimization: Lessons Learned from Aircraft Routing', *Computers & Operations Research*, **98**: 165–84.
- Medard, C. P., and Sawhney, N. (2007) 'Airline Crew Scheduling from Planning to Operations', *European Journal of Operational Research*, **183**: 1013–27.
- Muter, İ. et al. (2013) 'Solving a Robust Airline Crew Pairing Problem with Column Generation', *Computers & Operations Research*, **40**: 815–30.
- Papadakos, N. (2009) 'Integrated Airline Scheduling', *Computers & Operations Research*, **36**: 176–95.
- Parmentier, A., and Meunier, F. (2020) 'Aircraft Routing and Crew Pairing: Updated Algorithms at Air France', *Omega*, **93**: 102073.
- Pilla, V. L. et al. (2008) 'A Statistical Computer Experiments Approach to Airline Fleet Assignment', *IIE Transactions*, **40**: 524–37.
- Pilla, V. L. et al. (2012) 'A Multivariate Adaptive Regression Splines Cutting Plane Approach for Solving a Two-Stage Stochastic Programming Fleet Assignment Model', *European Journal of Operational Research*, **216**: 162–71.
- Pita, J. P., Adler, N., and Antunes, A. P. (2014) 'Socially-Oriented Flight Scheduling and Fleet Assignment Model with an Application to Norway', *Transportation Research Part B: Methodological*, **61**: 17–32.
- Rexing, B. et al. (2000) 'Airline Fleet Assignment with Time Windows', *Transportation Science*, **34**: 1–20.
- Rosenberger, J. M., Johnson, E. L., and Nemhauser, G. L. (2004) 'A Robust Fleet-Assignment Model with Hub Isolation and Short Cycles', *Transportation Science*, **38**: 357–68.
- Rushmeier, R. A., and Kontogiorgis, S. A. (1997) 'Advances in the Optimization of Airline Fleet Assignment', *Transportation Science*, **31**: 159–69.
- Ruther, S. et al. (2016) 'Integrated Aircraft Routing, Crew Pairing, and Tail Assignment: Branch-and-Price with Many Pricing Problems', *Transportation Science*, **51**: 177–95.
- Saddoune, M., Desaulniers, G., and Soumis, F. (2013) 'Aircraft Pairings with Possible Repetitions of the Same Flight Number', *Computers & Operations Research*, **40**: 805–14.
- Saddoune, M. et al. (2012) 'Integrated Airline Crew Pairing and Crew Assignment by Dynamic Constraint Aggregation', *Transportation Science*, **46**: 39–55.
- Safaei, N., and Jardine, A. K. (2018) 'Aircraft Routing with Generalized Maintenance Constraints', *Omega*, **80**: 111–22.
- Şafak, Ö., Çavuş, Ö., and Aktürk, M. S. (2018) 'Multi-Stage Airline Scheduling Problem with Stochastic Passenger Demand and Non-cruise Times', *Transportation Research Part B: Methodological*, **114**: 39–67.
- Salazar-González, J.-J. (2014) 'Approaches to Solve the Fleet-Assignment, Aircraft-Routing, Crew-Pairing and Crew-Rostering Problems of a Regional Carrier', *Omega*, **43**: 71–82.
- Sandhu, R., and Klabjan, D. (2007) 'Integrated Airline Fleeting and Crew-Pairing Decisions', *Operations Research*, **55**: 439–56.
- Sarac, A., Batta, R., and Rump, C. M. (2006) 'A Branch-and-Price Approach for Operational Aircraft Maintenance Routing', *European Journal of Operational Research*, **175**: 1850–69.
- Schaefer, A. J. et al. (2005) 'Airline Crew Scheduling under Uncertainty', *Transportation Science*, **39**: 340–8.
- Shao, S., Sherali, H. D., and Haouari, M. (2017) 'A Novel Model and Decomposition Approach for the Integrated Airline Fleet Assignment, Aircraft Routing, and Crew Pairing Problem', *Transportation Science*, **51**: 233–49.
- Sherali, H., Bish, E., and Zhu, X. (2006) 'Airline Fleet Assignment Concepts, Models, and Algorithms', *European Journal of Operational Research*, **172**: 1–30.

- Sherali, H. D., Bae, K.-H., and Haouari, M. (2010) 'Integrated Airline Schedule Design and Fleet Assignment: Polyhedral Analysis and Benders' Decomposition Approach', *INFORMS Journal on Computing*, **22**: 500–13.
- Sherali, H. D., Bae, K.-H., and Haouari, M. (2013) 'An Integrated Approach for Airline Flight Selection and Timing, Fleet Assignment, and Aircraft Routing', *Transportation Science*, **47**: 455–76.
- Sherali, H. D., and Zhu, X. (2008) 'Two-Stage Fleet Assignment Model Considering Stochastic Passenger Demands', *Operations Research*, **56**: 383–99.
- Smith, B. C., and Johnson, E. L. (2006) 'Robust Airline Fleet Assignment: Imposing Station Purity Using Station Decomposition', *Transportation Science*, **40**: 497–516.
- Sohoni, M., Lee, Y.-C., and Klabjan, D. (2011) 'Robust Airline Scheduling under Block-Time Uncertainty', *Transportation Science*, **45**: 451–64.
- Souai, N., and Teghem, J. (2009) 'Genetic Algorithm Based Approach for the Integrated Airline Crew-Pairing and Rostering Problem', *European Journal of Operational Research*, **199**: 674–83.
- Sriram, C., and Haghani, A. (2003) 'An Optimization Model for Aircraft Maintenance Scheduling and Re-Assignment', *Transportation Research Part A: Policy and Practice*, **37**: 29–48.
- Standfuss, T. et al. (2023) 'ANSPs in Turbulent Times - Uncovering the Impact of Demand Shocks on Efficiency Using the Malmquist Index', *Journal of the Air Transport Research Society*, **1**: 101–16.
- Subramanian, R. et al. (1994) 'Coldstart: Fleet Assignment at Delta Air Lines', *Interfaces*, **24**: 104–20.
- Sun, X., Wandelt, S., and Zhang, A. (2021) 'Technological and Educational Challenges towards Pandemic-Resilient Aviation', *Transport Policy*, **114**: 104–15.
- Sun, X., Wandelt, S., and Zhang, A. (2022) 'COVID-19 Pandemic and Air Transportation: Summary of Recent Research, Policy Consideration and Future Research Directions', *Transportation Research Interdisciplinary Perspectives*, **16**: 100718.
- Sun, X., Wandelt, S., and Zhang, A. (2023a) 'Aviation under the COVID-19 Pandemic: a Synopsis from Normalcy to Chaos and Back', *Journal of the Air Transport Research Society*, **1**: 136–51.
- Sun, X., Wandelt, S., and Zhang, A. (2023b) 'A Data-Driven Analysis of the Aviation Recovery from the COVID-19 Pandemic', *Journal of Air Transport Management*, **109**: 102401.
- Tahir, A. et al. (2021) 'An Improved Integral Column Generation Algorithm Using Machine Learning for Aircrew Pairing', *Transportation Science*, **55**: 1411–29.
- Talluri, K. T. (1998) 'The Four-Day Aircraft Maintenance Routing Problem', *Transportation Science*, **32**: 43–53.
- Wandelt, S., Sun, X., and Zhang, A. (2023) 'Towards Analyzing the Robustness of the Integrated Global Transportation Network Abstraction (IGTNA)', *Transportation Research Part A: Policy and Practice*, **178**: 103838.
- Wei, K., Vaze, V., and Jacquillat, A. (2020) 'Airline Timetable Development and Fleet Assignment Incorporating Passenger Choice', *Transportation Science*, **54**: 139–63.
- Weide, O., Ryan, D., and Ehrgott, M. (2010) 'An Iterative Approach to Robust and Integrated Aircraft Routing and Crew Scheduling', *Computers & Operations Research*, **37**: 833–44.
- Wen, X. et al. (2020) 'Robust Airline Crew Scheduling with Flight Flying Time Variability', *Transportation Research Part E: Logistics and Transportation Review*, **144**: 102132.
- Wen, X. et al. (2021) 'Airline Crew Scheduling: Models and Algorithms', *Transportation Research Part E: Logistics and Transportation Review*, **149**: 102304.
- Xu, Y., Wandelt, S., and Sun, X. (2021) 'Airline Integrated Robust Scheduling with a Variable Neighborhood Search Based Heuristic', *Transportation Research Part B: Methodological*, **149**: 181–203.
- Xu, Y., Wandelt, S., and Sun, X. (2023) 'A Distributionally Robust Optimization Approach for Airline Integrated Recovery under in-Flight Pandemic Transmission Risks', *Transportation Research Part C: Emerging Technologies*, **152**: 104188.
- Xu, Y. et al. (2024) 'Competitive Integrated Airline Schedule Design and Fleet Assignment', *European Journal of Operational Research*, **314**: 32–50.
- Yan, C., and Gbate, A. (2022) 'A Dynamic Model for Airline Fleeting and Scheduling'. Available at SSRN.
- Yan, C., and Kung, J. (2018) 'Robust Aircraft Routing', *Transportation Science*, **52**: 118–33.
- Yen, J. W., and Birge, J. R. (2006) 'A Stochastic Programming Approach to the Airline Crew Scheduling Problem', *Transportation Science*, **40**: 3–14.
- Zeighami, V., Saddoune, M., and Soumis, F. (2020) 'Alternating Lagrangian Decomposition for Integrated Airline Crew Scheduling Problem', *European Journal of Operational Research*, **287**: 211–24.
- Zhou, L. et al. (2020) 'Airline Planning and Scheduling: Models and Solution Methodologies', *Frontiers of Engineering Management*, **7**: 1–26.

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